The Politic

A WOMAN'S PLACE

Issue II
The Yale Journal of
Politics and Culture

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LETTER FROM THE EDITORS

Dear readers,

In *The Politic's* Issue II, our reporters explore how the personal and political are enmeshed through articles of local, national, and international significance. In "Uninvited Ears," Sovy Pham presents rigorous on-the-ground reporting on new audio surveillance systems in New Haven buses, examining the tension between personal privacy and public security. Andrew Xu analyzes the misrepresentation of ESG in "The Problem Isn't Sustainable Investing," showing how the politicization of sustainable finance has distorted perceptions of the burgeoning green investing movement.

In "Playing Catch-up," Samantha Moon traces how the systemic failures of COVID-era remote education have amplified existing disadvantages to learning. She demonstrates how a confluence of broad societal factors culminate in possibly permanent setbacks for individual students' educations. For our cover story, "A Woman's Place," Nicole Chen dives into the all-female presidential race in Mexico—looking at how a matchup between two women might shake up gender dynamics across the country. Phoenix Boggs examines climate change as a public health crisis in "Climate Fever," arguing that rising temperatures, natural disasters, and a host of other problems threaten to place unbearable pressure on global healthcare infrastructure. And in "Deserted Mothers," Nicole Manning investigates how the underfunding of hospitals in rural areas has endangered pregnant people, who are left without adequate obstetrics and postpartum resources. Her piece illuminates the deeply intimate effects of a failing healthcare system.

This issue's interview is with Yale Global Affairs professor and intelligence expert Ted Wittenstein YC '04, YLS '12. As he recalls how his college years imbued him with a desire to serve the country, our conversation with Wittenstein teases out the personal roots of his commitment to improving cybersecurity and international cooperation. Finally, in Issue II's photo essay, "Shenzhen Rewind," Linxi Cindy Zeng uses photography to forge a closer connection with the city she now calls home.

Putting this magazine together has been a great joy for the Managing Board, and we extend our highest gratitude to the brilliant reporters, designers, and editors that made its production possible. *The Politic* is a labor of love, and we're beyond eager to share this issue's pieces, which spotlight the individual stories underlying today's most pressing news.

Sincerely,
The 2023Managing Board

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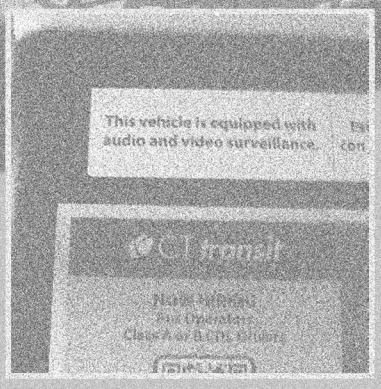
“Generally there aren’t limits on the government’s ability to put cameras and surveillance systems up on government and public property, be it a government building, a bus, or a street corner. That’s increasingly creating serious problems of pervasive surveillance in cities that are now densely populated with government cameras.”



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BY **SOVY PHAM**

The Debate over Audio Surveillance on New Haven Buses

ON A BRISK NOVEMBER EVENING in New Haven, the bus was on time. On a CT Transit bus, a young man clad in a fast food uniform was asleep. A middle-aged mother gripped tightly onto a stroller where her daughter fussed inside. Two teenage girls gossiped. An elderly man rode home from the doctor's office; a couple held their week's groceries in brown bags. As the bus lights flickered, passengers watched New Haven blink by.

None of them were aware of the small black half-spheres that were monitoring their every move. Without notice, a silent, uninvited audience tapped their conversations. On the bus, four cameras watched, listened, and stored the intimate details of their day to day lives.

Since August 1st, all New Haven city buses have had audio surveillance systems installed to capture both visual and voice recordings of patrons. On the four buses I rode through the city, there were no signs alerting the passengers that in addition to video surveillance, their conversations were being recorded.

Thomas Stringer is the General Manager of CT Transit, an organization owned and operated by the Connecticut Department of Transportation. As general manager, Stringer oversaw the surveillance policy implementation this past summer.

Alluding to the recent surveillance implementation, Stringer told *The Politic* "this was a missing piece in the safety program.Being new management and new leadership, we're identifying and addressing issues that should have been addressed many, many years ago, and bringing the agency up to the current standards of systems around the country."

Before joining CT Transit, Stringer worked in a number of other public transit agencies, including Cincinnati Metro, Jacksonville Transportation Authority, the Charlotte Area Transit System, and the Metropolitan Atlanta Rapid Transit Authority. He said the previous transit systems he's worked in have deployed audio surveillance systems on their buses.

"Every place I've been, we've had audio on the buses. So [its absence] was unusual, and an anomaly here," Stringer said.

Surveillance systems in public transportation primarily emerged as a post 9/11 artifice.

"After 9/11 there was a very significant swing towards the idea that more surveillance and undermining privacy would provide greater security," said Jake Laperruque, deputy director

of the Center for Democracy and Training's Security and Surveillance Project.

"The PATRIOT Act is a major example, especially section 215," according to Laperruque. Since its passage, section 215 the Patriot Act has made it substantially easier for the government to seize a large volume of records. Laperruque also pointed to a number of federal measures designed to procure personal data from civilians, including the "Terrorist Surveillance Program that President Bush unilaterally enacted, the augmented use of National Security Letters, and Section 702 of FISA."

With the creation of the Transportation Security Administration (TSA) under the U.S. Department of Homeland Security by President George W. Bush, all transportation systems in the United States were consolidated under a federal agency to execute improvements in security measures. Federal grants and funding from the Department of Homeland Security has allowed major metropolitan transit authorities across the country to increase their surveillance capacities.

"Over time, and especially because of the Snowden revelations and debate stemming from them, the attitude that there was always a tradeoff between privacy and security changed," Laperruque said.

With the expiration of the Bush-era surveillance laws, the USA Freedom Act was enacted in 2015 to limit bulk collection of telecommunication metadata on US citizens by federal agencies, including the National Security Agency.

Laperruque explained that the act was a major turning point for protecting privacy rights: "Congress voted by a large margin to outlaw bulk collection, and made a variety of other surveillance reforms, most notably requiring public disclosure of important FISA Court decisions, so new surveillance systems couldn't exist in secret."

The majority of cities that have deployed transit surveillance systems do not conduct them in secret, although passengers are not made explicitly aware of how the data is stored or utilized.

Major metropolitan transit systems did not deploy video or audio surveillance systems until the early 2010s. When San Francisco MTC rolled out the installations in 2012, there was pushback on the surveillance system with critics calling it "Big Brother-esque." The Maryland Transit Administration (MTA) began capturing both audio and video recordings in 2013. In an interview with *Government Technology* in March 2016, MTA Police Force Captain Chistropher Holland defended the system, stating, "it's a public setting on a public conveyance. On that same bus, anybody could be pulling out a cell phone and recording what you're doing."

Cities across the country have continued to quietly implement auditory surveillance systems alongside visual recordings, but the Connecticut Department of Transportation did not take such measures until this year.

When the announcement was first sent out to CT Transit employees, the Amalgamated Transit Union Local 281, which represents New Haven City bus drivers, immediately expressed concerns about their members' privacy rights. Though the policy was initially slated to take effect on July 1, 2023, CT Transit did not begin utilizing audio capacity on transit until August 1, 2023 due to union backlash, according to Stringer.

"I told them, I'll give you an additional 30 days to work that feeling out and work it out. And we did. But I told them on August 1, we would be activating the onboard recording audio as well as the video and it hasn't been, there's been no issue related to it," Stringer recounted.

Ralph Buccitti is the chairman of the Amalgamated Transit Union Local 281 chapter. He told *The Politic* that the increased surveillance measure " was a unilateral decision by management to implement the policy, and [they] never discussed with the union or negotiated to put it into the labor agreement."

Buccitti said that the other two ATU union groups, ATU Local 443, representing Stamford, and ATU Local 425, representing Hartford, are also fighting against CT Transit's audio surveillance system. The joint bargaining council has since filed a grievance procedure against CT Transit, arguing that the transit authority violated initial labor contract agreements between itself and the three ATU unions.

"We have now invoked arbitration to bring it to a third party in order to listen to both sides and make a decision on whether or not what stays or goes," Buccitti said.

Buccitti contends that the legal justification for the surveillance rollout is hard to navigate. "I'm not a lawyer. I mean, this is stuff we've looked up on the internet," he said. "But I can't get anybody to explain to me how it doesn't violate eavesdropping laws. You're listening to a conversation without consent, and I don't wish to be a party to it."

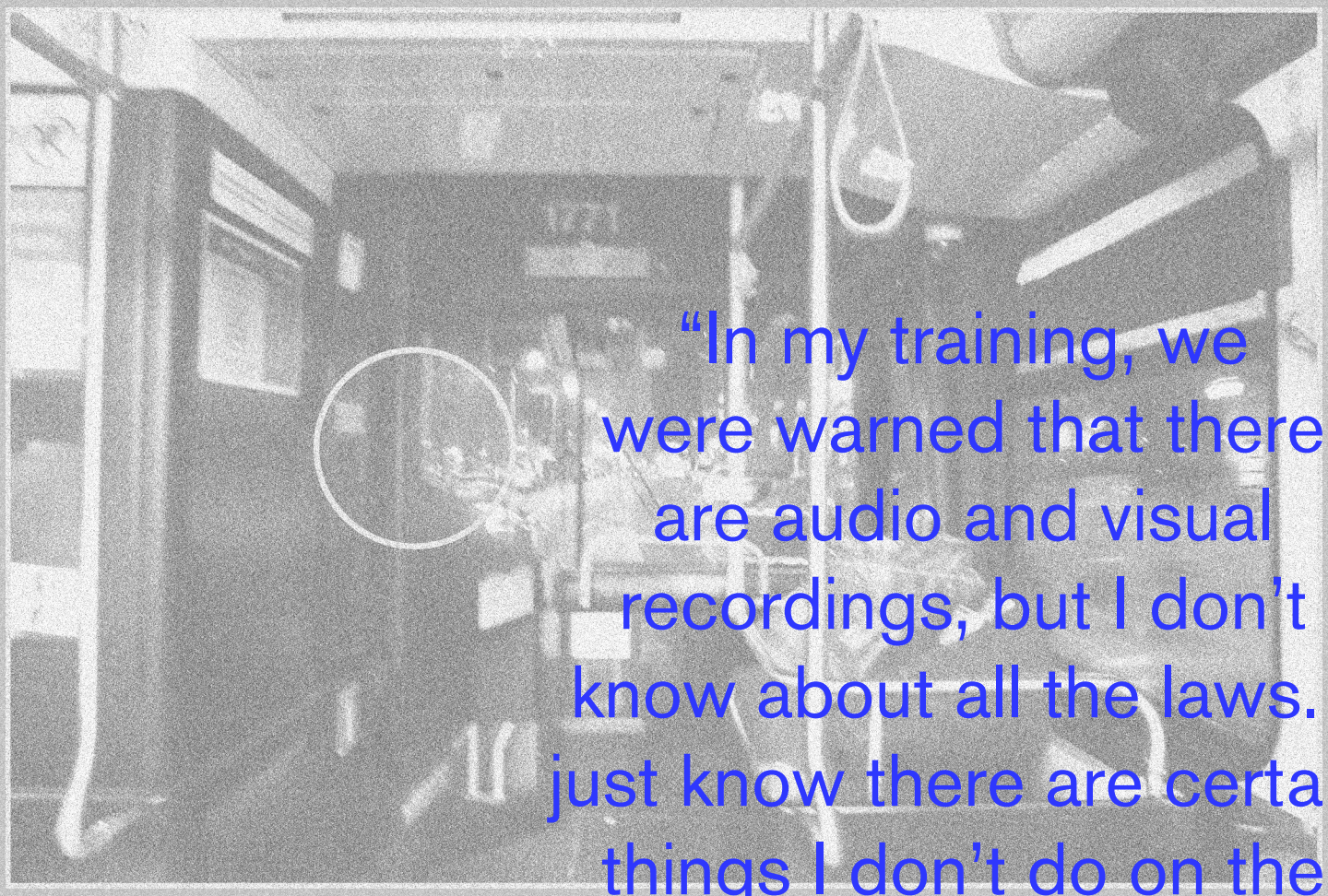
The union groups have also raised questions surrounding the efficacy of the audio surveillance system. "We haven't had anything, to my knowledge, outstanding that the audio would prove or disprove," said Buccitti.

Buccitti also raises concerns about labor violations. Under Connecticut labor laws, employees who work seven and a half or more consecutive hours must be given a break of at least 30 consecutive minutes for a meal. Buccitti explains that "for bus operators, when they're out on their route, they may have a few minutes of break time. They're entitled to their own personal space, but they can't even have a conversation without being constantly recorded."

Employers are allowed to deploy surveillance systems in their workplaces, but must inform employees if surveillance is used and what its purpose is. Additionally, surveillance should be used in public areas where employees do not have a reasonable expectation of privacy.

"When you are in a break area for a break time, like a lunch room in a work setting, you are not allowed to have audio or video recording," Buccitti said. "But the bus drivers are now being recorded in their own personal space during this short time window for their breaks."

Since the bus drivers' workplace is the bus itself, drivers only get breaks when all patrons are off of the bus. Without the ability to leave the bus during their shift, drivers are confined to being recorded 24/7.



"In my training, we were warned that there are audio and visual recordings, but I don't know about all the laws. I just know there are certain things I don't do on the bus, or say on the bus."

"People have brought up in the beginning the union's privacy concerns, but I tell them, there is no expectation of privacy in public spaces," Stringer said. "We are being recorded just going about our day throughout the city on the streets." He dismisses the idea that the CT Transit audio surveillance system could be used to parse out individual information about a particular bus driver, saying that nobody has time to process the video and audio taken on the buses.

Stringer also argues that the net positive of the audio recording outweighs any individual privacy concern, helping to control rowdy bus riders. He said the presence of the surveillance devices pushes riders to "conform to reasonable expectations" of etiquette.

LAST YEAR, GOVERNOR NED LAMONT signed the Connecticut Data and Privacy Act into law. The Act grants Connecticut residents rights over their personal data by establishing privacy protection standards for data controllers. Stringer maintains that the new policy does not violate CDPA.

"The Act doesn't apply to this because it's a public space," Stringer said. "Recognizing that all the buses have the notification saying that riding on this vehicle is subject to audio and visual recording, so there is no violation. Again, the reasonable expectation, you're in a public space, so therefore, you can't expect privacy."

Could these recordings be used against bus drivers for disciplinary measures? Could the accrued data be used to potentially spy or "mission creep" — government use of surveillance data in ways other than disclosed—on unsuspecting passengers?

"Generally there aren't limits on the government's ability to put cameras and surveillance systems up on government and public property, be it a government building, a bus, or a street corner. That's increasingly creating serious problems of pervasive surveillance in cities that are now densely populated with government cameras," Laperruque said.

Given technological limitations, authorities have historically relied on the argument that the recordings do not violate wiretapping laws, since the devices do not have facial recognition or biometric collection capacity.

When asked his opinion on the future of mass transit surveillance systems, Stringer explained that if new systems were installed, the data from bus rides could be stored, overwritten,

and distributed. While he concedes that the collected data could potentially be abused in the future, Stringer maintains that CT Transit is nowhere near that capability.

"With the amount of volume of recording that goes on a daily basis, nobody's listening to them. It's only pulled if there's an incident, and it rewrites every 30 days. So if something happened in September, it's not even there anymore," he said.

CHANDU GOGINENI COMMUTES TO HIS COLLEGE using New Haven city buses. When asked if he felt safer on the bus with the surveillance system, Gogineni said he didn't feel so.

But Gogineni still believes that the addition of the audio surveillance system is a net positive. "I don't mind anything about this," he said. "Since I work in the field of artificial intelligence, data is required. Facial recognition, geospatial mapping—we require data to work on these projects. This would be useful for that."

Laperruque pushed back on this. "I think facial recognition and remote biometric identifiers present the biggest threats. With automated identification tools like facial recognition, the government might soon be able to effortlessly track movements and activities."

Another bus driver, who requested to only be identified by his bus route, 6130, became a driver at CT Transit two months ago. He was born and raised in New Haven, and has been riding the bus since childhood.

"In my training, we were warned that there are audio and visual recordings, but I don't know about all the laws. I just know there are certain things I don't do on the bus, or say on the bus," the 6130 driver told *The Politic*.

The 6130 bus driver also did not believe that the audio recording system worked as a public safety measure, as Stringer claimed it set out to do. "It doesn't really prevent anyone from being belligerent." But the 6130 bus driver believed that the audio recordings could be used by law enforcement to resolve passenger or bus driver conflicts that may arise on board. "When it comes down to the law and they can come back and get the tape, it comes in handy," he said.

Another bus passenger, Marquis Faison Jr., feels ambivalent about the situation. Faison commutes on the bus daily to get to his workplace, Frank Pepe's.

"I think it's good for public safety. But I still think it's a little bit of an invasion of privacy," Faison mused. "What I talk about isn't everybody's business. But when you talk in public you kind of make it everybody's business. It doesn't mean that it needs to be recorded, though."

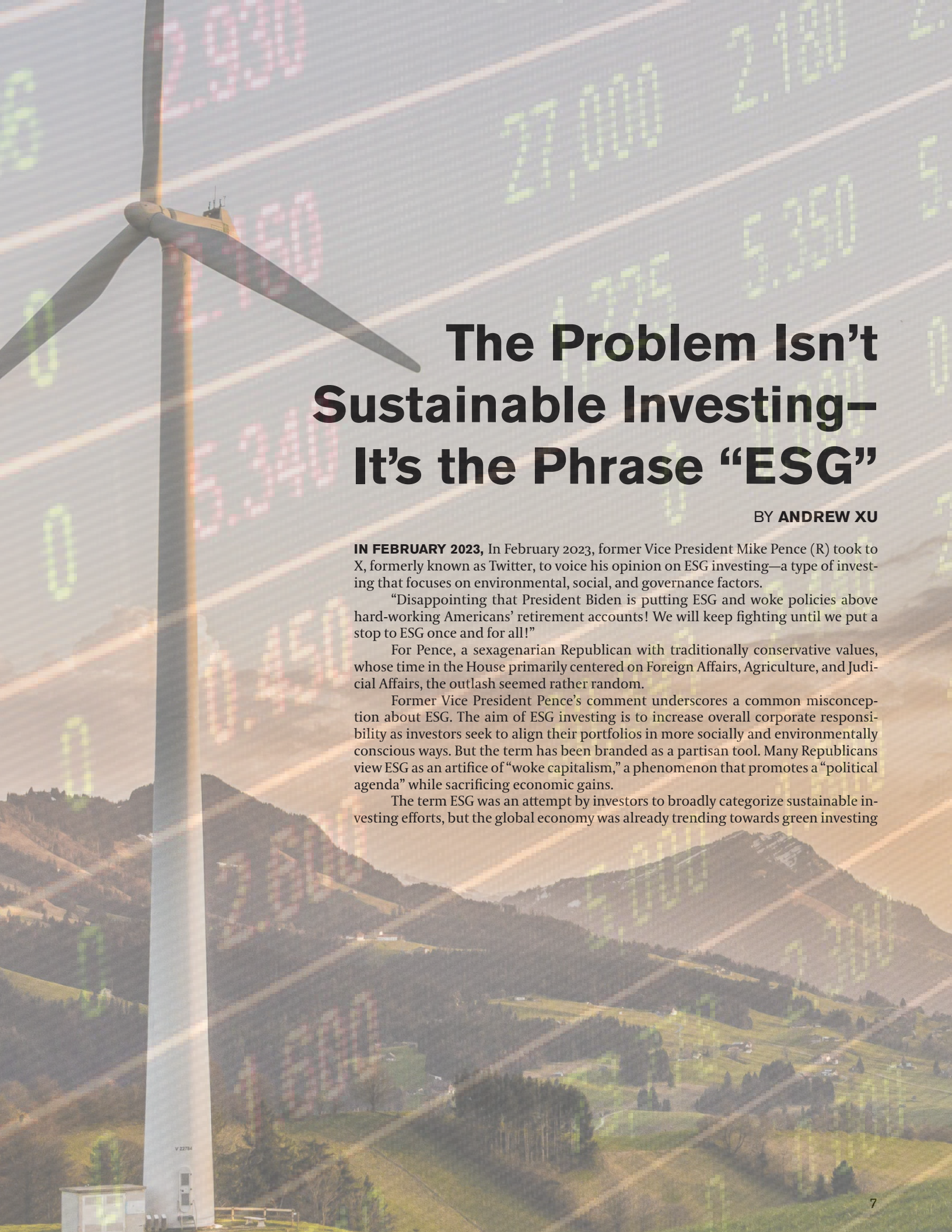
"What I talk about isn't everybody's business. But when you talk in public you kind of make it everybody's business. It doesn't mean that it needs to be recorded, though."

When asked if he had noticed a difference in passenger behavior, Faison said he didn't believe so.

"This is New Haven, Connecticut. There's no controlling the rowdiness on the bus. It's just not gonna happen. Are you going to have a bus monitor for adults?"

The majority of the bus drivers *The Politic* spoke to seemed to be accepting of the new changes. But Buccitti worries about the bus drivers' ability to openly speak out against the monitoring system. "When it comes time for the fight, most of the time, people don't want it. They're worried about being retaliated against, or being blacklisted."

But Buccitti remains optimistic on the unions' joint grievance to repeal the new surveillance measure. "We supposedly live in a democracy, but things are just the way that they are and people shrug their shoulders and accept that. Well, as a union leader, I don't believe in shrugging my shoulders. I believe in standing up and fighting."



The Problem Isn't Sustainable Investing– It's the Phrase "ESG"

BY **ANDREW XU**

IN FEBRUARY 2023, In February 2023, former Vice President Mike Pence (R) took to X, formerly known as Twitter, to voice his opinion on ESG investing—a type of investing that focuses on environmental, social, and governance factors.

"Disappointing that President Biden is putting ESG and woke policies above hard-working Americans' retirement accounts! We will keep fighting until we put a stop to ESG once and for all!"

For Pence, a sexagenarian Republican with traditionally conservative values, whose time in the House primarily centered on Foreign Affairs, Agriculture, and Judicial Affairs, the outlash seemed rather random.

Former Vice President Pence's comment underscores a common misconception about ESG. The aim of ESG investing is to increase overall corporate responsibility as investors seek to align their portfolios in more socially and environmentally conscious ways. But the term has been branded as a partisan tool. Many Republicans view ESG as an artifice of "woke capitalism," a phenomenon that promotes a "political agenda" while sacrificing economic gains.

The term ESG was an attempt by investors to broadly categorize sustainable investing efforts, but the global economy was already trending towards green investing

before the phrase. According to Cary Krosinsky, a faculty member at Yale's Center for Business and the Environment, "[ESG] is sort of an oversimplification, which is necessary so that something can resonate with the general public, but a lot gets lost in translation." The bulk of ESG's publicity problem comes from data indicating that ESG funds result in less profit than comparable funds. But the data doesn't tell the full story.

ESG may be perceived by conservatives as a "woke" policy, but fundamentally, sustainable investing is about making smart economic decisions. If an investor cared about the environment over profit, they could invest in ESG funds. If they cared about both the environment and profit, they could invest in ESG funds. And if an investor only cared about profit, they could *still* invest in ESG funds. At the end of the day, regardless of the type of fund, profit can be found anywhere.

15 years ago, there was no political connotation to the phrase ESG. "ESG is kind of a uniquely investor focused term. Investors had already been considering environmental, social and governance factors as financial risks, so when they read sustainability reports, they were converting sustainability information into environmental, social, and governance risks into financial performance. They already had this terminology of ESG–it wasn't a branding thing," says Todd Cort, the Faculty Co-Director of the Yale Center for Business and the Environment.

TO UNDERSTAND HOW this technical term has made its way into mainstream American political discourse, it is necessary to examine investment trends of the past decade. The term was coined in 2006 by the United Nations to require companies to include ESG criteria in their data to further the process of sustainability. This followed the Kyoto Protocol, the world's first legally binding climate treaty that took effect in 2005. Since then, ESG has grown into a mammoth industry: one measure estimates that ESG investment is globally valued at over 17.5 trillion U.S. dollars. According to Jason Eis, a partner at McKinsey & Company specializing in sustainable investing,"It's remarkable how much things have changed in the last five years. If you take a look at the balance of investment towards sustainable versus unsustainable, there's been a massive shift into the way people think about the future."

The increased focus on climate change led to radical changes. In 2017, 62% of ExxonMobile's shareholders voted for the world's largest gas and oil company to report the effects of climate change on business. As sustainable investing became more incentivized and traditional industries such as fossil fuels faced tough times, ESG numbers skyrocketed. And as it became a large focus on Wall Street, the public began to take notice.

Conservatives honed in on the data related to ESG. Analysis from market evaluations showed that ESG funds performed worse than a broad selection of funds from across the market, which then performed worse than only politically neutral companies. Conservative critics lambasted that investors were concerned with maximal profits, not environmental impact.

However, the data that their displeasure stems from is misleading. This isn't an issue of the analysis: ESG indices do perform worse. However, this is because ESG is a misfitting classification for sustainable investing. "There's a whole list of these [ESG] firms that are financially outperforming. So if you actually try to financially outperform and have a smart strategy that is possible," claims Krosinsky.

Why is this an issue? Much of the ESG index is a result of greenwashing or low-effort investing. Greenwashing occurs when investors claim they are taking environmental impact while investing in different funds. Low-effort investing occurs when the institution often does not care for returns as much as for their claims, so they do not screen their funds as well as they should to look for profit.

Greenwashing has become a prominent concern, with companies such as Volkswagen and Amazon receiving fines for exaggerating their environmental impact. This phenomenon has led to distorted ESG ratings. To combat this, the U.S. Securities and Exchange Commission (SEC) instituted a policy in October 2023 that set regulations for what can and cannot be labeled an ESG fund. According to Cort, "the purpose is to provide clarity around the the funds strategy, rather than the label." Although not a complete solution, regulations such as these will provide a clearer view of the return of optimal ESG strategies.

IT WILL STILL TAKE a while for ESG to leave the political spotlight, though. Cort says: "There are many barriers. [On] either side of politics, there's money, there's donors with interests who give money for a particular reason. They'll primary you if you don't follow their request."

Fundamentally, apart from donors, the political Left and Right have split opin-

Professor Tyler Wry at the University of Pennsylvania's Wharton School of Business says, **"this whole idea that ESG is somehow anti-capitalist is a cynical and inaccurate take."**

ions on ESG. According to Cort, "You can see where some of the political Right comes in and sees an SRI (socially responsible investment) or an impact fund, and says that's not fiduciary duty. And someone from the Left comes in seeing this ESG integrated fund where you're just trying to maximize returns by avoiding environmental litigation liability. That's greenwashing! But they're both called ESG, so they can use these radically different conclusions."

Companies are hesitant to invest in historically profitable industries, such as fossil fuels, because climate regulations are constricting them and they will inevitably dominate a smaller share of the market year after year. And strategies are aligning sustainability with profit.

Professor Tyler Wry at the University of Pennsylvania's Wharton School of Business says, "this whole idea that ESG is somehow anti-capitalist is a cynical and inaccurate take." ESG just recognizes the economic impact of factors such as climate change on potential companies to invest in, and chooses those companies that will likely have strong future returns in addition to current strong returns.

DESPITE EVERYTHING, ESG is only projected to grow more. According to a study by Deloitte, climate change is expected to be the biggest concern on investor's minds by 2024. Just this September, California introduced significant regulations over disclosing climate-related financial risks, which will help assist future ESG investing in one of the largest markets in the U.S.

Eis thinks that the role of ESG is undergoing a fundamental shift. "We've seen lots of fluctuations, but we've seen a lot of expectation and higher growth and sustainable oriented businesses." The U.S. Sustainable Investment Forum (SIF) estimated that around 13% of the current U.S. market of assets under management is ESG-related.

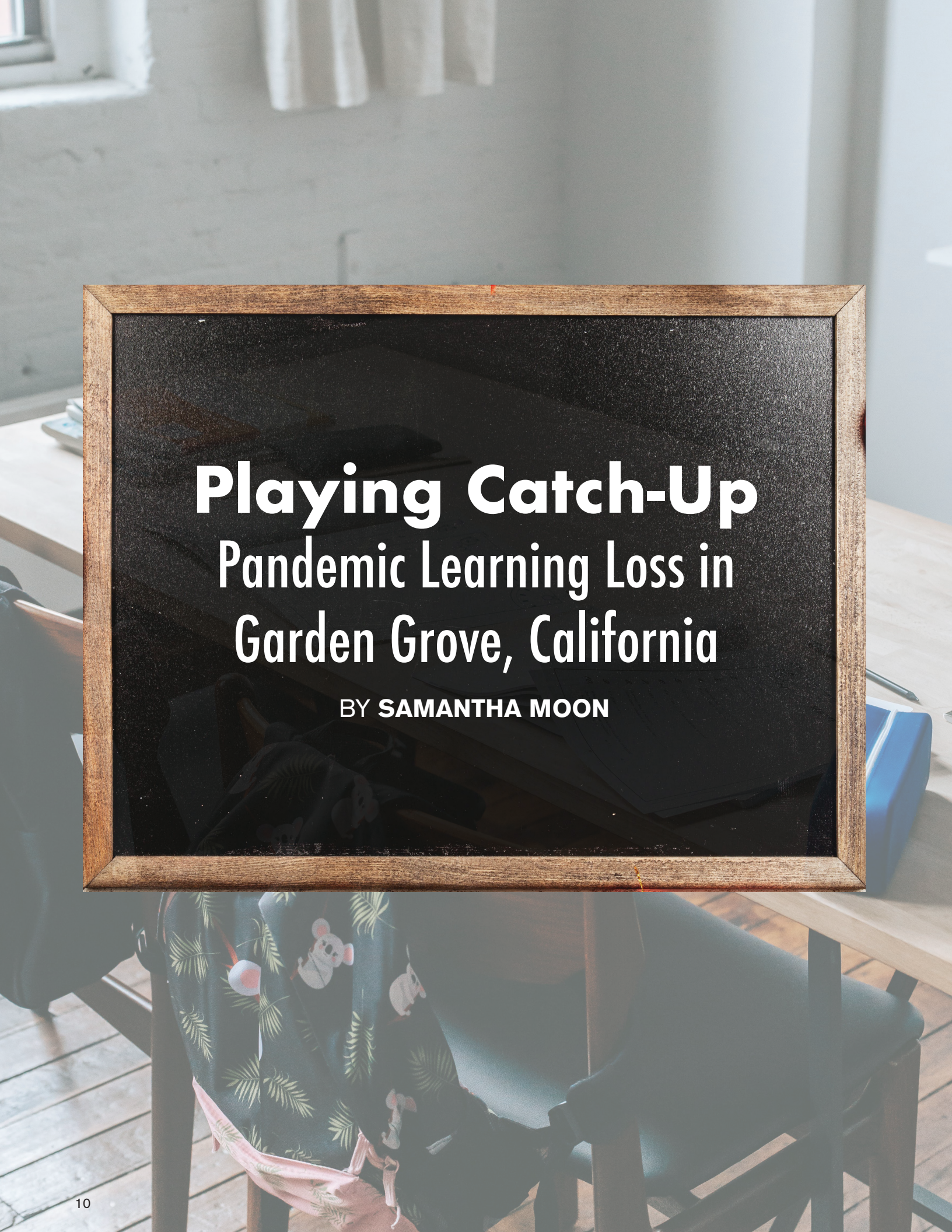
Investors seem to be bullish on this investment strategy that doesn't have great numbers.

"Overall, when you look at the evidence of how ESG funds perform, it tends to be a little bit lower right now. But then every time they refine ESG metrics, the link between doing this type of screening on the investments and the return of the fund gets better," says Wry. This is a problem with data, not a problem with investment. Smart investors will continue to be good at their job.

While ESG's politicization is not ideal, experts agree that this won't significantly impact its future.

According to Wry, "[Investors] are going to do some stuff that is socially responsible, they want to have a good image. They don't want to be lightning rods for criticism. But fundamentally, the purpose is to maximize profits."

They won't listen to the Right and abandon an area of profit because of misconceptions, but they also won't listen to the Left and create more social change for less profit. ESG is truly capitalistic.



Playing Catch-Up

Pandemic Learning Loss in Garden Grove, California

BY **SAMANTHA MOON**

**Note: The student "Joseph" is referred to by a pseudonym to protect his privacy as a minor.*

IN THE FALL OF 2021, Garden Grove's Sunnyside Elementary reopened its doors, returning to in-person learning after the COVID-19 pandemic shut down Southern California's public schools. At this time, Kellee Kim had just started teaching second grade at Sunnyside. She had previously been a teacher for kindergarten and transitional kindergarten, a grade level between pre-k and kindergarten. Ms. Kim knew the first year back in-person would be challenging. Kids would be behind. But the reality she faced was far more drastic than she ever imagined. "That year was the worst year in my 27 years of teaching," she told *The Politic*.

In the classroom, many of Ms. Kim's students struggled to perform at a second-grade level. They had difficulty spelling simple words and adding and subtracting double digit numbers. Beyond academics, even simple social skills—sharing, raising their hands—were entirely absent. These skills, which ordinarily would have been taught in the kindergarten classroom, could not be taught over a computer screen.

"My students were in dire need," Ms. Kim said. "They had no idea how to hold a conversation. They were behind academically. They struggled to write their names. Most of them couldn't even hold a pencil."

After two years of online learning, many of Ms. Kim's seven- and eight-year-olds lacked the foundational knowledge and skills that kindergarten and first grade normally provide. While they had all, ostensibly, attended Zoom school, family and household complications created a challenging learning environment for many students. According to Ms. Kim, even those students living in situations conducive to academic success absorbed frighteningly little while online.

"My students were in dire need. They had no idea how to hold a conversation. They were behind academically. They struggled to write their names."

Ms. Kim's situation isn't unique. Several years after emerging from isolation, it has become clear that students across the U.S. are struggling to recoup pandemic-related learning loss. Teachers and parents alike recognized that the quality of education and student engagement suffered in the transition to online school. Many publications have reported the pandemic's adverse effects on education. In particular, many articles argued that a student's socio-economic status largely determined their degree of learning loss during the pandemic.

In October 2022, Tom Kane, director of the Harvard Center for Education Policy Research, and Sean Reardon, a sociologist at the Stanford Educational Opportunity Project, created the education recovery scorecard, a national "report card" that tracked COVID-based academic achievement losses in U.S. public schools. According to the scorecard, learning losses were significantly higher in higher-poverty districts. The quarter of districts with the highest proportion of students receiving federal lunch subsidies lost two thirds of a grade level of learning, and students living in low-poverty districts lost nearly half a grade level.

In the United States, poverty and race are highly correlated. Relative to their total populations, vastly more people of color (POC) live in poverty

than white people. According to a study by George Farkas, a professor emeritus of education at University of California, Irvine, Black children tend to be less prepared for kindergarten than white children. Additionally, many Asian and Hispanic students have lower achievement levels because a high proportion of their parents lack English skills. The COVID pandemic exacerbated pre-existing racial inequalities in the education system.

During the height of the pandemic—in order to mitigate the harmful effects of COVID on their children's educations—many parents and guardians became active participants or educators themselves, hiring outside tutors or homeschooling. According to an article in *Frontiers in Education*, family involvement plays a crucial role in supporting student learning and well-being. However, not all families had the time or resources to dedicate to their child's education. In the U.S., the people with the fewest resources are disproportionately POC. "When you have a massive crisis, the worst effects end up being felt by the people with the least resources," Reardon said on PBS's NewsHour.

"Many Sunnyside families had to put education on the back burner in favor of survival."

Sunnyside Elementary, Ms. Kim's school, exemplifies this reality.

Sunnyside is a public school in Garden Grove, California—a city of 170,488 in Orange County. Garden Grove is not an affluent city. Approximately 13% of Garden Grove residents are "persons in poverty," meaning that their family's total income is below the U.S. Census Bureau's poverty threshold. The percentage of Garden Grove residents considered 'persons in poverty' is higher than that of the state of California and the U.S. as a whole.

Garden Grove is a diverse city with a large immigrant population. Over three quarters of Garden Grove residents are Asian or Latine. A large portion of the immigrant community comes from a Vietnamese heritage, with strong Korean, Mexican, Chinese, El Salvadoran communities as well. Roughly 44% of Garden Grove residents are classified as immigrants as opposed to California's one quarter, and two thirds of households speak a language other than English at home, which is well above the state average.

Sunnyside is very much a reflection of its city. Currently, 95% of students at Sunnyside are non-white, and as of the 2021-22 school year, 73.80% of Sunnyside students were classified as socioeconomically disadvantaged. Sunnyside is considered a Title One school and is part of a federal education program that supports low-income and high-poverty students through grants.

Four in ten students at Sunnyside are English learners. Ms. Kim estimates that a quarter of her students' parents either can't read and write in English or are barely doing so at an elementary level. These parents were unable to supplement their children's education and help them with their homework during the pandemic. Additionally, some non-English-speaking parents couldn't understand school announcements or written correspondence from teachers.

The living conditions of many students also complicated their online learning experiences. For families struggling financially, creating space for their child to participate in online learning was not the number one priority; putting food on the table was. Ms. Trcka, a first-grade teacher at Sunnyside witnessed, firsthand, this reality.

"Already living in multi-family homes or apartments, families were constantly forced on top of each other," Ms. Trcka said. "They did not have reliable WiFi, nor did they have their own spaces to learn. Too many interruptions made for inadequate absorption of materials. Many Sunnyside families had to put education on the back burner in favor of survival."

In short, their ability to learn was compromised.

Dr. Christina Cipriano, an assistant professor in the Child Study Center at the Yale School of Medicine, offered an additional lens through which to view COVID-era learning disruptions: the impacts on students' availability to learn. Dr. Cipriano defines 'availability to learn' in terms of the social, emotional, and affective components of things such as stress, anxiety, safety, responsibilities at home, and experiences of discrimination or prejudice.

"During the first phase of COVID, with loss of income, loss of jobs, and loss of life compounding, particularly in pockets of the US where we saw disproportionate rates of deaths among families of color, we're having a kind of disproportionate loss of life," she said. "Those factors can contribute both to how students show up in school, at the onset."

Ms. Trcka sees the reflection of this loss every day in her first graders. "As we all know, the pandemic has left a huge emotional scar for many people. There was so much sickness and loss that families are still healing and dealing with these losses. Kids are now coming to school with major behavioral issues that have not been diagnosed for a treatment plan, and Gen-ed teachers are expected to help them cope. It's almost too much to ask of us," she said.

Ms. Cho, a teacher at the neighboring Allen Elementary, had a different experience of online learning altogether. "While families of students in low-income areas were on survival mode, my school community continued to thrive due to the extra support they received from their family and resources," Ms. Cho said.

Allen Elementary is located four miles south of Sunnyside—a roughly 14-minute drive. Despite its proximity, Allen's neighborhood is demographically very different from Sunnyside's. While also a part of the Garden Grove Unified School District, Allen is not technically in Garden Grove, but Fountain Valley, a smaller, whiter, and more affluent city.

The percentage of persons in poverty in Fountain Valley is 8.3%, five percent less than in Garden Grove. Additionally, Asian and Latine people—the dominant POC groups in Fountain Valley—comprise just half of Fountain Valley's population, and the percentage of immigrants is 31.4%; recall that for Garden Grove, these numbers are over three quarters and 43.9%, respectively.

Overall, the demographics of Fountain Valley were more conducive to success-

ful online learning experiences than those of Garden Grove. This is reflected in the contrasts between Allen and Sunnyside. At Allen, only 40.50% of students are socioeconomically disadvantaged as opposed to Sunnyside's three quarters. On average, Allen parents are wealthier, better employed, and better educated than Sunnyside parents. They have the time and resources to help their children. Additionally, a greater proportion of Allen parents and students speak English—over 80% of Allen students are English speakers. "Unlike most of my friends that taught at schools with a lower socio-economic demographic, I had a lot of parents that sat with their child during class time to help," Ms. Cho said.

The 2021-22 California state test scores for Allen and Sunnyside are a testament to the impact of city demographics on children's quality of education. In the 2021-22 school year, Allen students greatly outperformed Sunnyside students in mathematics and English language arts (ELA). Over 86% of Allen students met or exceeded state standards in both subjects. At Sunnyside, just 58.66% of students met or exceeded standards in mathematics, while 65.24% did so in ELA.

This imbalance existed long before the pandemic; it's representative of a greater cycle in which societal advantages breed success, while disadvantages breed stagnation. Since 2014—which is the earliest year for which the Department of Education contains data—roughly 20-30% more Allen students than Sunnyside students have exceeded state standards in math and ELA each year.

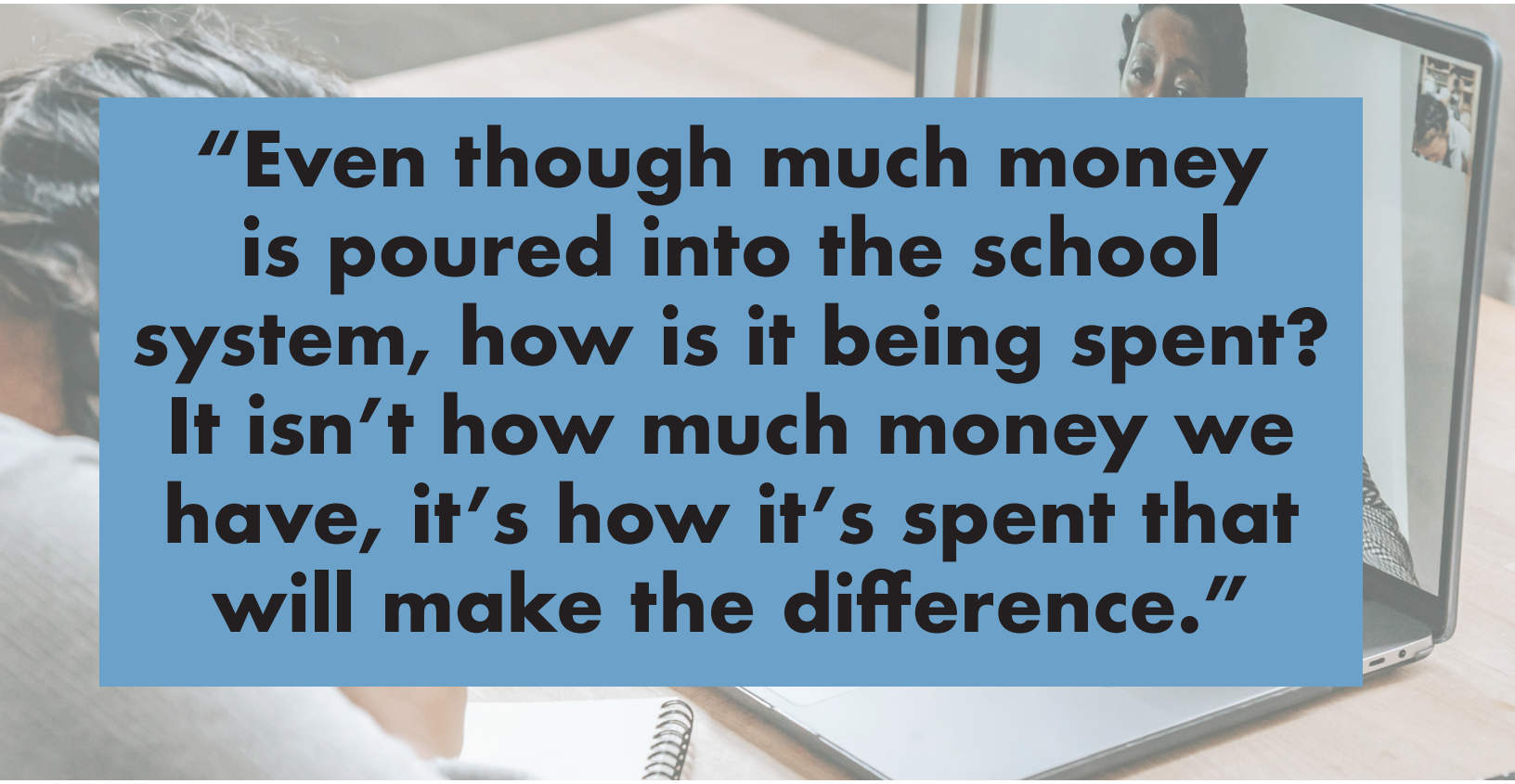
Garden Grove and Fountain Valley—thus, Sunnyside and Allen—are a microcosm of the way city demographics impact education and student outcomes nationwide. When students lack an educational support system at home, as do many socioeconomically disadvantaged students, who are often also POC or children of immigrants, it's easier to fall behind. It's also harder to catch up.

What results is the reinforcement of cycles of poor education and poverty. According to Opportunity Atlas—a platform that uses census data to map the social mobility outcomes of children into adulthood based on their parents' incomes—in Garden Grove and surrounding areas, higher parent incomes lead to better child outcomes. Lower parent incomes, on the other hand, translate to children having lower employment rates, lower individual incomes, lower household incomes, and higher incarceration rates.

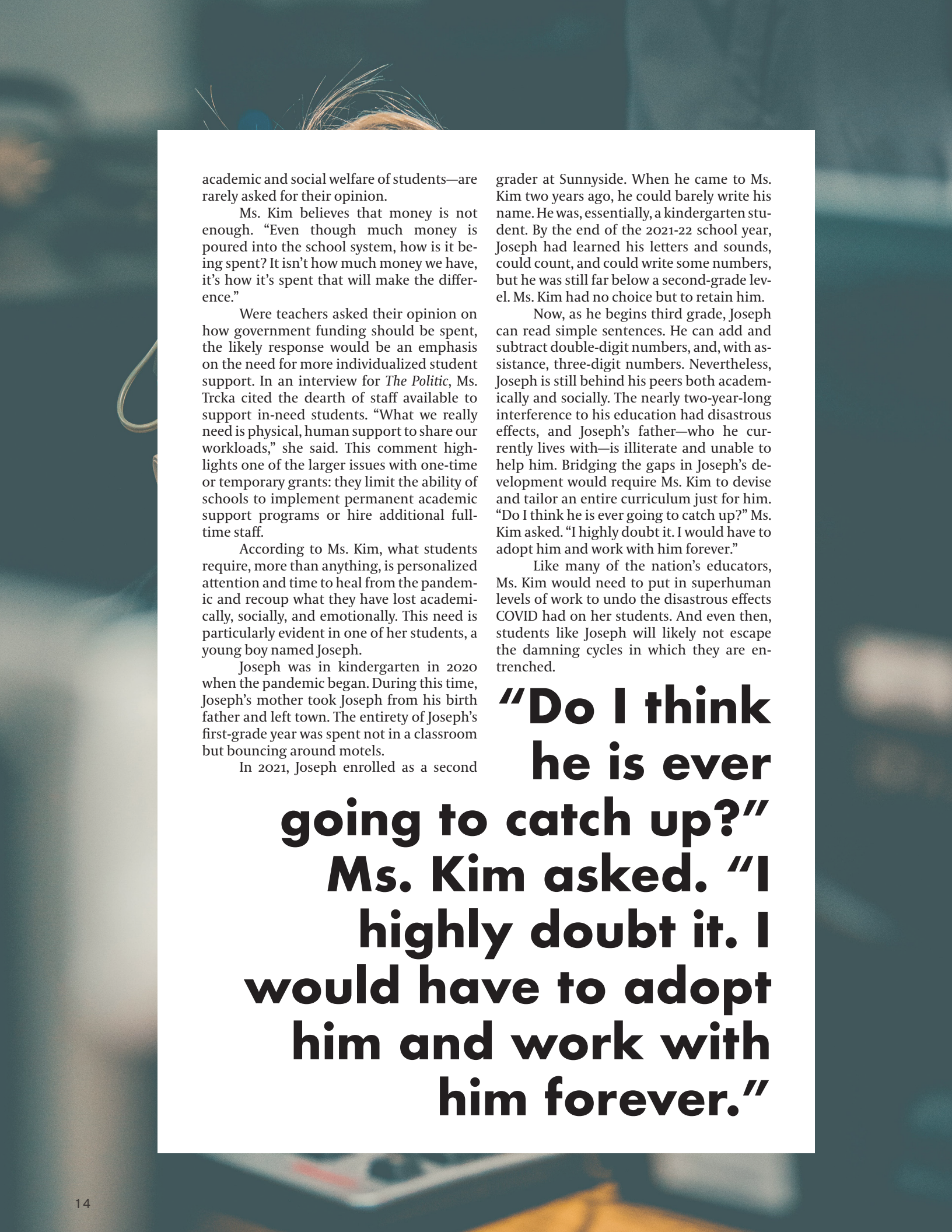
As many immigrant and POC families—those of Sunnyside, included—have discovered, poverty is an incredibly difficult cycle to break. When the education system fails already disadvantaged children, as it did during the pandemic, their hope of breaking the cycle decreases exponentially. Children who face cumulative, systemic disadvantages are effectively predisposed to fail.

Still, state and local governments are trying to bridge the educational gap created by the coronavirus pandemic. Through various COVID-relief grants and aid programs, California in particular has taken the first steps towards addressing COVID-related learning loss. In 2021, the California Department of Education created the $4.6 billion Expanded Learning Opportunity Grant, which allows schools to expend funds for the following categories: extending instructional learning time, accelerating progress to close learning gaps, integrated pupil supports, community learning hubs, supports for credit deficient pupils, additional academic services, and training for school staff.

While grants like these are certainly important, their categories of use are broad, and there is not much oversight to ensure that the funds have a positive impact on students' academic recovery. Often, it is school districts or county offices of education who make decisions about the specific uses of such funding. Teachers and school support staff—those on the ground and actually responsible for the day-to-day



"Even though much money is poured into the school system, how is it being spent? It isn't how much money we have, it's how it's spent that will make the difference."



academic and social welfare of students—are rarely asked for their opinion.

Ms. Kim believes that money is not enough. "Even though much money is poured into the school system, how is it being spent? It isn't how much money we have, it's how it's spent that will make the difference."

Were teachers asked their opinion on how government funding should be spent, the likely response would be an emphasis on the need for more individualized student support. In an interview for *The Politic*, Ms. Trcka cited the dearth of staff available to support in-need students. "What we really need is physical, human support to share our workloads," she said. This comment highlights one of the larger issues with one-time or temporary grants: they limit the ability of schools to implement permanent academic support programs or hire additional full-time staff.

According to Ms. Kim, what students require, more than anything, is personalized attention and time to heal from the pandemic and recoup what they have lost academically, socially, and emotionally. This need is particularly evident in one of her students, a young boy named Joseph.

Joseph was in kindergarten in 2020 when the pandemic began. During this time, Joseph's mother took Joseph from his birth father and left town. The entirety of Joseph's first-grade year was spent not in a classroom but bouncing around motels.

In 2021, Joseph enrolled as a second grader at Sunnyside. When he came to Ms. Kim two years ago, he could barely write his name. He was, essentially, a kindergarten student. By the end of the 2021-22 school year, Joseph had learned his letters and sounds, could count, and could write some numbers, but he was still far below a second-grade level. Ms. Kim had no choice but to retain him.

Now, as he begins third grade, Joseph can read simple sentences. He can add and subtract double-digit numbers, and, with assistance, three-digit numbers. Nevertheless, Joseph is still behind his peers both academically and socially. The nearly two-year-long interference to his education had disastrous effects, and Joseph's father—who he currently lives with—is illiterate and unable to help him. Bridging the gaps in Joseph's development would require Ms. Kim to devise and tailor an entire curriculum just for him. "Do I think he is ever going to catch up?" Ms. Kim asked. "I highly doubt it. I would have to adopt him and work with him forever."

Like many of the nation's educators, Ms. Kim would need to put in superhuman levels of work to undo the disastrous effects COVID had on her students. And even then, students like Joseph will likely not escape the damning cycles in which they are entrenched.

"Do I think he is ever going to catch up?" Ms. Kim asked. "I highly doubt it. I would have to adopt him and work with him forever."

A WOMAN'S PLACE

BY **NICOLE CHEN**

THE RACE FOR MEXICO'S FIRST FEMALE PRESIDENT

CLASSROOMS ARE NOW tense encampments for student protestors at Universidad Nacional Autónoma de México (UNAM), one of the most prestigious universities in Mexico. Marisa Belausteguigoitia, a full-time lecturer at UNAM whose research focuses on collective young feminist protestors, shared that close to 150,000 students are now seeking refuge in these makeshift barracks. As Belausteguigoitia described, "they are mainly protesting gender issues, protesting harassment, and also transsexual issues—anything that smells of sexuality or gender." The students' zeal has left them without classes for several months.

While these protests persist, the broader political landscape in Mexico is shifting: next year, the country will elect its first woman president. The emergence of two female presidential candidates—Claudia Sheinbaum and Xóchitl Gálvez—signifies a pivotal moment, reflecting a potential evolution of baseline attitudes towards gender norms. With the prospect of a woman assuming the highest office, there is hope that female leadership could be the catalyst needed to close Mexico's gender gaps. The student occupations at UNAM are not isolated incidents; they are rooted in a long-standing tradition of advocacy for gender equality, an ideal that is far from the current reality in Mexico.

According to the Center for Strategic and International Studies, there are ten femicides per day in Mexico—a rate that has doubled in the last five years. This violence has driven continued demonstrations. "The Day of Dead Women" protests, for example, occur each year after the nation's traditional celebrations for the Day of the Dead. All across Mexico, protestors carry crosses with the names and pictures of femicide victims—ranging from central Mexico City to the southern, rural state of Chiapas, to share their grievances about violence against women.

Diana Medina is responsible for institutional strengthening at a Mexican feminist fund known as Fondo Semillas. She explained in an interview with The Politic that "Mexico is a very centralized country in Latin America, so most protests tend to happen in Mexico City. The fact that this is happening in other states shows how the gender movement is breaking historical ways of centering power."

The 2018 election of Andrés Manuel López Obrador, popularly known as AMLO, initially provided hope for political change to female voters, as his new cabinet consisted of nine women and 10 men, which is "the closest gender differential ever seen in a Mexican president's cabinet." Subsequently, though, his reinforcement of the state's repressive institutions—such as the military—have left feminists disillusioned. Popular discontent with his presidency has led him to brand feminists as reactionaries.

In a leaked document from the Secretary of National Defense outlining "enemies of the state" who pose a threat to the International Felipe Ángeles Airport in Mexico City, López Obrador's administration identified "feminist collectives" among terrorist groups and cartels. Concurrently, the Mexican War on Drugs has allowed narco-politics to govern the administration, fueling a 236% increase in gender violence over the past four years.

In recent years, Mexico's gender protests have grown more intense. Protests have swelled from tens to hundreds of thousands of participants, and violence at demonstrations has increased. Some women have brought blowtorches, sledgehammers, and baseball bats to the National Palace, the presidential residence, ready for a confrontation with government forces.

This surge in gender activism has its origins in traditional notions of gender roles and machismo, a misogynist form of exaggerated masculinity that is still present in the feminist conversation today. Andrea Isita is the Digital Communications Coordinator of PSYDEH, a Mexican grassroots non-profit organization that invests in rural and Indigenous women. Her organization took a trip to a rural municipality, where PSYDEH's efforts have offered women the opportunity to earn a second income.

Isita said that the strongest resistance to her efforts came from the men of the municipality: "the idea is that men are the breadwinners, and even if the breadwinners are not home, the women cannot leave because they have their families and homes to tend to. They are very traditional in that sense, so it is very complex."

This pattern continued when, on another occasion, a PSYDEH volunteer visited a community to teach women about the benefits of Google Lens. The workshops were boycotted by all the men of the community. They declared that "the Internet was the worst for women because they would neglect their home duties and lose their morals as a result." Other civil organizations have witnessed similar pushback. At Fondo Semillas, Medina remarked seeing an increase in requests for grant funding from groups that work to fight violence, particularly for survivors of femicides.

While civil organizations have shown determination and achieved considerable success in addressing gender-related issues, the Mexican government has been struggling to pull its weight. A 2023 report from the International Federation for Human Rights revealed that in the past 60 years, there have been no convictions for female disappearances and just one arrest warrant in Guerrero. Such governmental neglect is largely a product of the profound discrimination women face when bringing forward these claims, often facing blame for the crimes committed against them.

Despite the federal government's general apathy and incompetence in combating gender-based violence, there have been vigorous efforts to promote female representation in government. Gender quotas were introduced in the early 2000s, and as of 2021 50% of lawmakers in Mexico's lower house of Congress are women.

Indeed, the gendered shift in Mexican politics has been extraordinary. As Jesús Javier Silva-Herzog Márquez, a professor and researcher of politics and political theory at Tecnológico de Monterrey, noted, "the President of the Supreme Court is a woman, the President of the House of Representatives is a woman, the President of the Senate is a woman. This is a very fast change that we see in the government. This is a gender revolution in Mexican politics."

Even though access to these offices is important, some are questioning the utility of established gender quotas. Georgina Flores is the Operations and Knowledge Management Coordinator of Aúna, a Mexican-based platform that promotes women in political leadership. Flores argues that Mexico needs to take a step further than just gender parity politics because even if women win political office, they are often overlooked or absorbed into the existing patriarchal hierarchy.

"Gender quotas can sometimes be seen as a barrier, wherein women run but once they are in office, they are not actually the people making the decision," Flores said. "Most of the time, women do not have the backing from political parties or from big networks like men have."

But presidential candidates Sheinbaum and Gálvez stand out. Large political networks and backing from Mexico's largest political parties are underpinning these two womens' competing bids for the presidency in the upcoming 2024 elections. The prospect of a female president is an unprecedented step towards genuine representation for women in the highest office of the country.

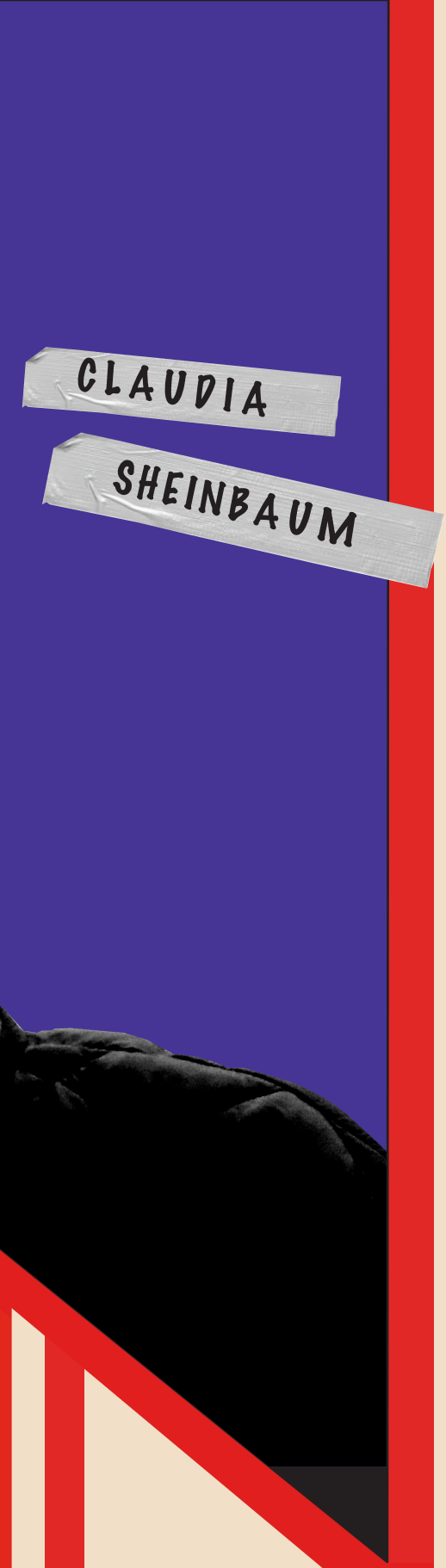
Sheinbaum, the favored candidate with an 18-point lead as of October 31st, comes from Mexico's left-wing party, MORENA. She has the backing of the popular incumbent president López Obrador, an endorsement that has tremendously shaped her campaign. According to Flores, "Claudia basically started her campaign several years ago by being so close to AMLO [López Obrador]. He legitimizes her, and I think that has a lot of weight in who she is." Sheinbaum's biggest challenge thus far has been distinguishing herself as an independent candidate, as opposed to a puppet of López Obrador.

Sheinbaum is a Jewish-Mexican politician and academic who earned a bachelor's degree in physics, a master's degree in energy engineering, and a doctorate in environmental engineering from UNAM. She has long been active in politics, first serving as the Secretary of the Environment for Mexico City in 2000, then as the mayor of Tlalpan, and most recently as the mayor of Mexico City.

If elected, Sheinbaum has announced that she would continue to pursue many of López Obrador's policy goals. Specifically, she claims she would avoid overhauling fiscal reforms, pursue social welfare policies, and uphold policies for Mexico's national energy companies, PEMEX and CFE. In line with her expertise, she has called for investment in better recycling plants and public transportation. In regards to her stances on crime, Sheinbaum supports a stronger National Guard presence to make arrests and keep the military in airports in retaliation to the ongoing War on Drugs.

Sheinbaum's record on gender issues has been uneven. During her tenure as mayor, she "was not light-handed in using police force against feminist protests," said Mayte López, an author and Spanish professor at Yale University. Belausteguigoitia offered a similar account, arguing that Sheinbaum was "very strategic and calculated, making decisions to specifically put policewomen at the frontlines of the protests."

Consequently, Sheinbaum was able to accuse protestors of deriding their own principles by attacking policewomen. After an abortion protest in September 2020, Sheinbaum claimed that she supported the cause but condemned the violence of the protestors and did not offer any justifications for the police force that she had released.



On the other side of the race is Gálvez, who is supported by a three-party coalition known as *Frente Amplio por México*. This coalition consists of the National Action Party (PAN), a right-wing party, the Institutional Revolutionary Party (PRI), a center-right party, and the Party of the Democratic Revolution (PRD), a left-leaning party.

Though the support from a coalition of varying ideologies for Gálvez may seem surprising at first, Silva-Herzog Márquez explained the reasons underlying this tension. "AMLO [López Obrador] is basically a populist leader that has been able to assemble a very popular government. And this antagonistic idea of politics has pushed the old enemies, the PRI, PAN, and PRD to create this new coalition to fight the strong political movement that is MORENA," said Silva-Herzog Márquez. Indeed, Flores added that this coalition represents a "desperate decision to be in alliance against MORENA," but is not entirely novel, as the parties have run as a coalition in previous local elections.

Before entering politics, Gálvez received her degree in computer engineering from UNAM and founded two technology companies. Afterwards, she served as a general director for the National Institute of Indigenous Peoples under then-president Vicente Fox, was the mayor of borough Miguel Hidalgo, and served most recently as a senator in the Mexican Congress.

Unlike Sheinbaum, who shies away from speaking about her Jewish heritage, Gálvez's Indigenous background is central to her political identity. However, some question Gálvez's alliance with a conservative party like PAN. Isita draws a comparison to María de Jesús Patricio Martínez, also known as Marichuy, who was a previous Indigenous presidential candidate.

"There is a big difference between Marichuy and Xóchitl. They are both Indigenous-identifying, but they come from different worlds," said Isita. "Marichuy has been a leader of her community and politically involved with the locals. On the other side, Xóchitl is wearing a costume: she came from a low-income family and has risen up, but she is perceived differently because she rubs elbows with the richest of the rich."

In terms of policy, Gálvez's platform emphasizes greater support for businesses and industries. She hopes to change the status quo by reforming PEMEX and promoting private investment in the energy sector. Though an advocate for some of López Obrador's social welfare policies, Gálvez hopes to further expand government aid programs that create opportunities instead of short-term cash handouts. With reference to her stance on crime, she rebuffed López Obrador's slogan of "hugs, not bullets" and declared that "there will be no hugs" for those contributing to the War on Drugs.

Her record on feminist policies has also been mixed. On the International Day of Indigenous Women, Gálvez demanded justice for Indigenous women who suffer heightened rates of femicides. But at the same time, she has been incongruous in her support of policies like abortion. Before the presidential race, she claimed that she fully supported access to abortions, but after being chosen to represent Frente Amplio, her stance shifted: "my position is that I lead a Broad Front where different positions fit, and I will be respectful of each one. As Frente Amplio, I am obliged to respect different visions," she told *El Financiero*. Her inconsistency in standing by her statements obscures her policy platform and erodes trust in her intentions.

Though these candidates come from different sides of the aisle, they have a lot in common. They share support for similar social welfare policies and a scientific and technical perspective, which stands in "contrast to the politics of AMLO [López Obrador], where policies are very macho and impulsive," said Silva-Herzog Márquez. He elaborated that these candidates are "team builders and don't feel weakened by being surrounded by smart, experienced persons, which is something that is contrasting with the current President, a very impulsive, aggressive, polarizing figure."

Both candidates have found themselves similarly reluctant to tread outside of their respective political party structures. Many of their proposals are continuations of existing policies created by past male presidents. However, their unique experiences as women may offer fresh insights. Currently, however, they both present a somewhat ambiguous stance on gender politics, making it difficult to gauge their beliefs.

Some have argued that the two candidates' genders did not have a predominant role in the party's selection. Silva-Herzog Márquez noted that Sheinbaum "was clearly the most popular politician in the official structure of MORENA and had the most visible political position in the country as a successful mayor of Mexico City." To Xóchitl's bid, he said that she "was the figure that reacted brilliantly to the provocations of the president. And at a time where the opposition didn't have any strong political figures that could bridge the differences between these coalition parties, Xóchitl Gálvez was the person that would be an experienced politician, not clearly identified to be with one of the contenders, but somehow above all ideologies."

Sheinbaum and Gálvez's gender has created unique challenges and opportunities for their candidacies. Belausteguigoitia offered a distinct perspective with her use of philosopher Judith Butler's performance framework, which posits that gender is an act of performance shaped by cultural scripts and societal norms. Belausteguigoitia argued that within the political arena here, "there are different protocols, scenes, and infinitely exploitable ways to play the woman card. And the moment is fantastic for playing those cards because the public has learned to expand what a woman is." Her argument, emphasizing the value of what she terms a "productive fiction for politics," suggests a potential strategic use of gender in campaigning.

In contrast to arguments based on biology or culture to justify women in positions of power, Belausteguigoitia asserted that "these two women can wisely and productively play the 'woman card,' the idea that women are empathetic, caring, are better administrators, and look more at the unevenness of society." In short, Belausteguigoitia highlighted how the gender of these candidates will significantly shape their campaign strategies.

While leveraging their gender can offer tactical advantages for these two candidates, the intersection of politics and prevailing machismo norms may pose challenges to the candidates' public image.

"When talking to the common man, there is all this bias from people just as educated as I am," Isita said. "They will say stupid shit like 'women are too sensitive' and 'they do not know anything.' We are a long way from a fair race, and likeability is particularly complex for a woman in a political landscape."

Silva-Herzog Márquez offered a similar perspective. "I think we still have a very strong misogynistic culture. If you turn on the TV, you might feel insulted with many stereotypes … the perspective of a woman is basically a housewife used as a sexy decoration of a TV program," he said.

López's assessment was more blunt: "there is a feeling that women are somehow disposable in a lot of parts of the country, which deeply worries me."

The degree to which the candidates' gender will shape their policies remains open to debate. As López stated, "the fact that they are women does not mean that they have a feminist agenda." Medina echoed her claim, adding that the research she has seen demonstrates that women reaching high positions of power does not mean they are going to have a feminist perspective.

Andrik García, a junior in Yale College, was born and raised in northern Mexico. He stated, "Mexico has other problems that cannot be solved by just having a female candidate. Problems are institutional, and no one president regardless of their gender, or identity can necessarily solve all the problems."

These implications take on a broader dimension when considering how the gender of a politician can affect Mexico's international image. Silva-Herzog Márquez noted that López Obrador has been absent in the global arena, so a new president can present opportunities to step up on the world stage. Silva-Herzog Márquez further noted that "in Latin American terms, Mexico will not be the first to have a woman president, but I do think that that is something that is obviously here to stay... you are not seeing just these two figures alone in Mexican politics, you are seeing something today that I was not used to seeing as a young guy."

Such novelty comes with fears for the future. Flores remarked that she's afraid that these women will be absorbed into the machismo culture. "I fear that when women are elected for this big, unprecedented office, they have to act bigger or have stronger policies towards war in a way to compensate and overcorrect the use of violence," she said. "Violence in Mexico is big enough as it is, and I would not want a woman to try to stand her ground in a violent way to prove herself."

Silva-Herzog Márquez worried that after "allowing women to be within these walls of men, we will feel that we did good and that we are justified in closing the door again after doing our part," echoing author Malcom Gladwell's idea of moral licensing.

But there are also hopes. According to García, "it is always optimistic to look at Mexico and see massive progress in terms of finally electing a female president who can be an inspiration to the future generations of women."

Medina and Isita both agreed that the feminist movement is the hottest movement on the agenda right now, and both expressed hope that this momentum can push the two candidates to pursue real feminist policies and challenge misogynistic norms.

This election marks a transformative moment in the intersection of gender and politics. Electing a female president brings gender to the forefront of political discourse in Mexico. Women in politics tend to experience heightened scrutiny, but this is also an opportunity for female leaders to sever the deep-rooted association between power and machismo.

Sheinbaum and Gálvez's success is not merely an independent achievement, but the result of a collaborative effort to reshape Mexico's political realm. In the words of Belausteguigoitia, "the way you occupy power as a woman is a collective thing. And if we hold onto the promise of womanhood, we have a future."

The Climate-Driven Infectious Diseases

A GLOBAL FEVER

Threatening Worldwide Health Systems

BY **PHOENIX BOGGS**

IN THE BEGINNING OF THE 19TH CENTURY, *New Orleans was one of the United States' wealthiest and largest cities. Its port shipped the produce of the American interior to worldwide markets, and wealthy merchants walked the streets in the latest European fashions. But amid this growth, a mysterious plague stalked the people of the Big Easy. Doctors remarked of a highly infectious and deadly fever that surged in the summer and retreated in the winter. Those afflicted suffered from mild fevers and headaches that quickly progressed into yellow, sunken skin and bleeding from the eyes or mouth. The victims' yellow complexions quickly earned the disease its notorious name: yellow fever.*

Outbreaks of yellow fever were an annual feature of life in 19th-century New Orleans. Some medical authorities attributed the outbreaks to "miasma," or unclean air, while others blamed the city's antiquated sanitation system. In one particularly deadly year, 20,000 people were killed by the disease in a conflagration that spread from New Orleans throughout the Mississippi River valley.

It was not until the turn of the century that officials realized what had caused the fatal outbreaks: Aedes aegypti *mosquitoes, which hitched rides to the city on merchant ships and thrived in the stagnant water of New Orleans' cisterns and gutters. A public health effort was mounted to overhaul the city's sanitation system and remove the mosquitoes' breeding grounds, and the epidemic gradually abated. The last New Orleans yellow fever outbreak occurred in 1905, and there has never been a major U.S. yellow fever epidemic since.*

IN OCTOBER OF 2023, the New England Journal of Medicine published an article summarizing an alarming academic consensus: there is danger for a potential resurgence of yellow fever in the United States, particularly in the southeast states. Scientists warn that "suitable conditions for future epidemics" of yellow fever are becoming prevalent in the U.S. again.

"If you look closely at the most recent scientific literature, you'll see that there is now some pretty serious concern about the return of yellow fever to the United States," Laurie Garrett told *The Politic*. Garrett is a journalist and infectious disease expert who won the Pulitzer Prize in 1996 for her work chronicling the Ebola virus outbreak in Zaire. She has spent her career sounding the alarm about the rising risk of infectious disease transmission worldwide.

Almost every aspect of modern human society—globalization, travel, urbanization—has increased the worldwide risk of pandemics and infectious diseases. But for the United States, the greatest risk factor for yellow fever is not urban overcrowding or travel. It's climate change.

Scientists and public health experts have long known that climate change would have disastrous effects on human health. But our understanding of the link between climate change and infectious disease, specifically, is more recent.

"Back in the 90s when I was writing *The Coming Plague*, the science was still pretty primitive. People were grasping at straws to figure out what was changing in the ecology," Garrett said. "By the time I wrote *Betrayal of Trust* [published in 2000], it was becoming much clearer. When I was working on my third book [2011], the relationship between climate change and disease was becoming terribly and painfully obvious."

Aedes aegypti and other mosquito species that spread disease thrive in warm, humid environments, which is why many mosquito-borne diseases are endemic to South America, Africa, and Southeast Asia. But as the world warms—global temperatures have risen by as much as 1.3 degrees Celsius from their preindustrial levels—the suitable habitat range for these mosquitoes is expanding.

In 19th century New Orleans, yellow fever arrived periodically on merchant ships coming from farther south. Once established, it ran rampant in New Orleans' summer climate but subsided every year when temperatures fell. Moving forward, scientists warn that southeastern U.S. cities will become warm enough for endemic transmission of the virus, even without the introduction of new travel cases from farther south.

Yellow fever is far from the only example. Cases of dengue fever, another mosquito-transmitted disease, have risen almost eight-fold since 2000. In October, the World Health Organization's chief scientist warned that climate change will cause dengue cases to proliferate in parts of the US and Europe, places with no previous history of the disease.

Yale Professor Dr. Robert Dubrow is an epidemiology specialist and the Faculty Director of the Yale Center on Climate Change and Health. He sees significant potential for the expansion of vector-borne diseases due to climate change.

"If the world doesn't take climate change seriously, then I think there's real potential for the arbovirus infections that are associated with *Aedes aegypti* and *Aedes albopictus* mosquitoes [to expand]," Dubrow told *The Politic*. "Those are the two types that transmit dengue, Zika, yellow fever, and chikungunya. So far those haven't had much transmission if any in the United States, but I could see that changing in the future."

And when vectors expand to territories without immunity, the risk of widespread infection is heightened.

"That's why we saw huge outbreaks of Zika and chikungunya when they were first introduced into the Americas, because the population was all naive and nobody had immunity," Dr. Scott Weaver said. Weaver is Chair of the Department of Microbiology and Immunology at the University of Texas and an expert in vector-borne diseases.

Mosquitoes are not the only vectors whose expanding ranges are becoming a cause for concern. As animals migrate in search of cooler habitats, they bring their parasites—particularly

ticks—with them.

"We are already looking at vast expansion in tick habitats and tick-borne diseases—everything from Lyme disease to Crimean Congo hemorrhagic fever," Garrett said. "Partly, it's driven by the movement of host animals as they change their territorial range due to climate impacts or deforestation impacts."

In 2020, the Centers for Disease Control and Prevention (CDC) released a report chronicling the spread of vector-borne disease in the United States. This report found that cases of vector-borne diseases more than doubled between 2004 and 2019 "as the ranges of vectors have expanded and the number of emerging pathogens have multiplied."

However, it can be hard to connect individual cases to this trend. Dr. Neha Pathak, a primary care provider who writes about health for *Yale Climate Connections* and other publications, notes that it is difficult to determine whether a patient's illness is the result of the broader pattern of climate-driven infectious disease or not.

"On the individual scale, it's very difficult for clinicians to say that, for example, because of climate change, you now have Lyme disease," Pathak said. "But I think on a population level, we can counsel our patients that there is a higher risk for Lyme disease spreading to our region, and because of that, we need them to be aware of the risk."

As climate change brings infectious diseases to new territories, it also worsens their current impact in endemic territories. Rising heat and humidity help vectors like mosquitos thrive and expand in their old environments. In 2022, *The Lancet* published a report stating that although dengue had always been endemic to South America, "the climate suitability for dengue transmission reached its highest level in recent years." The report found that, over the last thirty years, dengue transmission in South America increased by an estimated 35.3%.

It is not only vector-borne diseases that are surging in the global south. Natural disasters such as hurricanes, wildfires, and floods are becoming more and more frequent as the climate warms: the Ecological Threat Register found that the number of global natural disasters increased tenfold between 1960 and 2020. In 2022, natural disasters displaced 12.5 million people in South Asia who were forced to leave their homes and travel long distances to safer inland zones. These natural disasters wreak havoc upon health systems and create a perfect environment for the unchecked spread of infectious disease.

Natural disasters wreak havoc upon health systems and create a perfect environment for the unchecked spread of infectious disease.

In 2015, microbiology and immunology journal *Virulence* published a study crystallizing the relationship between climate disasters, displacement, and infectious disease. The study concluded that as climate change creates more and more refugees in coastal and equatorial regions, new diseases are being brought to new places. As they flee, refugees will travel routes containing diseases that are completely unfamiliar to

their immune systems. In the future, vulnerable populations and refugees will have to confront not only natural disasters but also rampant infection.

Of course, climate disasters also wreak havoc upon public health systems themselves. They exhaust and overextend local health agencies' resources and remove access to clean water, ample food, and adequate sanitation.

On October 25, 2023, Hurricane Otis slammed into Acapulco, Mexico. The hurricane's aftermath saw collapsed buildings and mangled sewer pipes, causing waste to flood the streets. Residents were forced to drink water from flood runoff, rivers, and puddles.

Disaster experts predict that as a result of dirty water, rampant sewage leaks, and insufficient government response, as many as 8,500 people in Acapulco could die of infections in the coming months. Giorgio Franyuti, CEO of emergency aid organization Medical Impact, commented to Courthouse News Service that the Acapulco government was not "prepared or equipped to adequately respond to the crisis."

It is clear that climate change presents new struggles for global infectious disease prevention systems that are already beleaguered by other risk factors.

"We're not helpless in the face of climate change. But we have to work even harder than we would if there were no climate change in order to control vector-borne diseases," Dubrow said. "The next pandemic could happen in two years, or it could happen in 25 years, or it could happen in 50 years. What we do know with really high confidence is that there likely will be another pandemic coming. We have to be forever vigilant against pathogens."

An important aspect of this vigilance is the production and dissemination of vaccinations. Vaccine technology is advancing rapidly, and many of the infectious diseases we are battling today already have good vaccines. Since the 1930s, a highly effective yellow fever vaccine has been widely available. While health authorities do not currently vaccinate most Americans against yellow fever, the vaccine's existence means that future efforts to counteract yellow fever will not start at square one.

Vaccines are not the only measure that we have against infectious diseases. New public health measures are already being put into practice in many places in response to novel infectious disease threats. In the United States, state departments of health are reviving and strengthening programs to track, surveil, and manage these infectious diseases. Dr. Elizabeth Borrero, an infectious disease specialist who has worked in Puerto Rico, Louisiana, and Florida (all places vulnerable to vector-borne disease due to their warm, humid climates), feels that state health departments are prepared to counter new infectious disease threats.

> "We know with really high confidence that there will likely be another pandemic coming."

"I would anticipate further outbreaks as these vectors expand. But I think we have the tools to monitor new disease threats," Borrero said. "State health departments are doing a really good job at keeping those things at bay. Especially in Florida, I think you would have seen a lot more outbreaks over the course of the last 10 years if it had not been for the trigger surveillance and monitoring of the state health department."

The Florida Department of Health (FDOH) employs several different disease surveillance programs, including online reporting systems for healthcare providers. But perhaps its most creative surveillance system is the "sentinel chicken" program. Across Florida, strategically placed coops filled with "sentinel chickens" act as an early public health warning system. Every week, mosquito control teams draw the chickens' blood to test for vector-borne diseases. By monitoring the chickens' health, the FDOH can focus its attention on the areas of greatest concern.

While efforts like the FDOH's sentinel chicken program are essential, there is certainly room to strengthen public health systems. Public health underfunding is a chronic problem in the United States. The budget of the U.S. Centers for Disease Control and Prevention (CDC), the country's leading public health agency, has been reduced by about half since 2002. Funding for the Hospital Preparedness Program, the country's primary source of federal funding for emergency health preparation, was reduced by nearly two-

thirds over a similar period.

It is not only U.S. health systems that have experienced budget problems. The World Health Organization (WHO) is widely considered to be grossly underfunded. On their website, the WHO states that in recent decades contributions of member countries have declined considerably, leaving the organization seeking voluntary donations from private donors to fund their health initiatives. In 2023, independent health policy source KFF reported that the WHO was trying to change its funding model in an attempt to make up for these budget gaps.

Chronic underfunding has real consequences for global health and safety. When emergencies occur, public health systems must confront high demand with limited resources. Dubrow notes that even after the worldwide COVID-19 pandemic, public health still remains under-prioritized and underfunded.

"Unfortunately, right now, we haven't learned the lessons from COVID-19 in terms of increasing public health capacity and preparedness," Dubrow said.

Dubrow suggests that efforts to prevent and manage future pandemics should focus on strengthening and empowering global public health systems, especially the World Health Organization.

"Strengthening global public health infrastructure could really make us all safer. In an ideal world, I would see the World Health Organization being more prominent and having more resources."

Pathak agrees that strengthening large-scale public health systems is crucial. But she also underscores the importance of training doctors to respond to these new threats on a patient-by-patient basis.

"We should think: what do we need to be doing that's different from before to protect our patients from the worst health impacts? I think that most clinicians need to become more climate-aware and expand the list of potential pathogens we think about so that we don't miss something that impacts the health of the patient in front of us," Pathak said.

In 19th-century New Orleans, yellow fever forced scientists and public officials to confront their city's public health weaknesses. Once they identified the cause of the fever, they drained cisterns and cleaned gutters until they expelled *Aedes aegypti*.

In the face of a world reshaped by climate change, public health systems will again be forced to adapt. As in 19th century New Orleans, the new threats could result in more robust, fortified, and capable public health systems. But for that to happen, current systems need more resources.

Climate change-driven disease could devastate global health. It could also be a catalyst for the creation of the strategies and tools we need to protect it.

DESERTED MOTHERS

IN A RECENT VIDEO released by March of Dimes, a pregnant mother named Claudia declares that she will likely give birth in a car. The nearest hospital is a thirty minute-drive from her home. Claudia recalls giving birth to her first child on a farm, her second in a car, and how she made it to the hospital to deliver her third just in time. Like so many others in Texas' Ward County, Claudia lacks close, convenient access to a physician or gynecologist in her area to provide obstetrics care.

BY **NICOLE MANNING**

89 *obstetrics units have closed in U.S rural hospitals between 2015 and 2019*

Her story is not the exception, but largely the norm for people from rural areas of the United States. According to the American Hospital Association, 89 obstetrics units have closed in U.S rural hospitals between 2015 and 2019. More than half of rural hospitals, already few and far in between, did not offer obstetrics care in 2021. Women in rural areas have all different challenges and needs, but have collectively experienced the effects of hospital closures due to underfunding and lack of care.

Ashley Stoneburner is a data science manager at March of Dimes, a nonprofit organization that works to improve health outcomes for mothers and babies. Stoneburner told *The Politic* that today, "there are 5.6 million women who live in counties with no or limited access to maternity care services." March of Dimes defines maternity care deserts as areas where there is limited or no access to birthing hospitals, birth centers offering obstetric care, or obstetrics providers. For those who live in maternity care deserts, "some rural residents have to travel as far as almost two hours to reach the nearest hospital with obstetrics services. That was the outlier, but that is the substantial burden," said Dr. Peiyin Hung, a professor at the University of South Carolina who specializes in geographic disparities in healthcare.

Lack of obstetrics care is just one of the issues pregnant people face in rural areas. "More than 20% of rural residents do not have the internet to access a lot of resources or know how to leverage existing community resources," said Dr. Hung. "Only looking at the most underserved communities leaves behind a lot of disadvantaged populations that have, on average, some resources—but who are still outliers compared to other communities."

As maternity care deserts continue to expand in rural areas, the maternal mortality rate has risen.

Maternal mortality refers to the death of a birthing person due to pregnancy-related health issues during childbirth or within 42 days postpartum, as defined

by the World Health Organization and the Center for Disease Control. From 2011 to 2013, the maternal mortality rate was 17 deaths per 100,000 births in the United States. Even though the U.S. has the highest healthcare spending per person in the world, the maternal mortality rate rose to 32.9 deaths per 100,000 live births by 2021. Black and American Indian/Native Alaskan (AI/AN) women are among the most vulnerable to complications—before, during, and after pregnancy. Notably, the maternal mortality rate for Black women is 69.9 per 100,000 births, and AI/AN women are 4.5 times as likely than non-Hispanic white women to die from their first delivery.

more than half of rural hospitals did not offer obstetrics care in 2021.

March of Dimes is currently partnering with Surgo Venture, a privately funded action tank, to measure the Maternal Vulnerability Index, which identifies where U.S. residents are most vulnerable to poor maternal health outcomes. The data from this report shows that people in states with a majority of rural counties—Alabama, Mississippi, Georgia, Texas, Arizona, Arkansas, and Oklahoma—are the ones most susceptible to maternal mortality.

Maternal deaths can be prevented if a woman meets with an obstetrician early and receives regular care. This not only improves the chances of a healthy pregnancy but also helps prevent common risks women face during and after birth, including hypertension, heart disease, stroke, and preeclampsia. Yet this support is not a reality for women who cannot quickly access obstetrics care or other resources efficiently.

Even programs such as Critical Access Hospitals, created by the Centers for Medicare and Medicaid Services (CMS) in response to over 100 rural hospital closures in the 1980s, do not include obstetrics care in their services. These Medicare-run institutions, which are either nonprofit or public, are intended to provide care in rural areas that are 15-to-35 miles from the nearest hospital or medical facility. These hospitals have at-most 25 acute care inpatient beds and do not offer care past 96 hours. Although this program has relieved some healthcare burdens in rural communities, it still leaves behind vulnerable populations of pregnant people.

Maternity care deserts arise mainly as hospitals struggle with staffing shortages, low patient volume, reimbursement, and funding to cover the cost of providing care. According to Christine Morton, PhD, a medical sociologist and author, "40 to 45% of all obstetric births are covered by Medicaid." But hospitals serving patients with Medicaid coverage often receive lower reimbursement rates, Morton continued. "Unfortunately, you have this perfect storm—where you have the most vulnerable women and other pregnant people who are in these low-resource areas and do not have access to care."

Many hospitals' leading concern is to generate greater revenue. "Hospitals, even if they are not for profit, are looking to increase margins of profitability, and obstetrics is not seen as a revenue generator in healthcare—unlike the NICUs, [which] generates a lot of income," Morton said.

In 1965, the U.S. federal government established Medicaid to improve health coverage for low-income people in all fifty states and U.S territories. The program plays a critical role in rural areas where families may not have jobs that provide healthcare or are underinsured, making it difficult to cover the care they need. While Medicaid is a vital resource for many women and families in rural areas, it does not pay or reimburse hospitals or clinics as much as private insurance does. The program is additionally limited because it was unequally expanded under the Affordable Care Act (ACA), which provided states the option to expand Medicaid coverage to

"some rural residents have to travel as far as almost two hours to reach the nearest hospital with obstetrics services."

childless adults and people who originally earned too much to qualify for Medicaid in the past. Due to a Supreme Court ruling in 2012, not all states are required to expand Medicaid, because of the increase in cost to cover a larger group of people. Now, the security of even having insurance may be taken away from many rural families. Legislation passed in 2022 that allocated billions of dollars to Medicaid expansion, in return for limiting states from dropping people from Medicaid rolls, will end next year.

When speaking to *The Politic* Usha Ranji, associate director for Women's Health Policy at KFF, explained, "one thing I hear from clinicians is that patient volume is really important for maintaining high quality care. In some communities, particularly the most rural communities, populations tend to be small. So it can be really hard to maintain the volume that you need to really maintain high quality, good clinical care."

Since 2020, one in five healthcare workers have quit their jobs

The more practice a physician receives in maternal care, the better they can serve and identify certain needs of women in these rural communities who already face so many risks. In 2016, the University of Wisconsin opened the nation's first—and so far only—program for OB-GYN residency in rural areas. Laura McDowell, MD, became the first resident to graduate from the program, in 2021. She explained to NBC News in an interview that she was inspired to enter the program because of her desire to combat inadequacies in rural healthcare. "Women shouldn't have to think twice about getting good quality health care in their small rural town," McDowell said.

McDowell's work is a part of a movement to boost recruitment of doctors to rural areas, especially as hospitals around the country struggle with staffing shortages. Since 2020, one in five healthcare workers have quit their jobs, while 47% of healthcare workers plan to leave their jobs by 2025. "One strategy we are starting to see is more federal dollars going towards supporting more training of healthcare professionals in rural areas," said Ranji. It can be hard to draw people there, but if you start building the workforce from people who already live in these communities, then they are more likely to stay." To address the staffing shortages afflicting hospitals in rural areas requires the effective reallocation of healthcare funds, in order to pay physicians more and stop punishing patients for being low-income.

The struggles in maternal healthcare extend to the healthcare mothers receive—or do not receive—after giving birth. Postpartum, or what is sometimes considered "the fourth trimester," is a vital time for mothers to receive care. The recent rise in postpartum depression and suicidality makes postpartum care especially crucial.

The ACOG advises expecting mothers to meet with their obstetrician several times during a period of 12 weeks after giving birth. However, many of these women living in rural areas likely will not have adequate postpartum care for more than sixty days, in states that have not adopted Medicaid expansion under ACA.

Sarah Benatar is a principal research associate for the Urban Institute who emphasized long-term maternal care in an interview with *The Politic*. "A lot of the issues we observe in terms of the care [received by] pregnant individuals, especially low-income pregnant people, is really just

fight. We all have a role to play in prioritizing U.S. Everyone can do their part so that all moms ossible start, right in their local communities."

[lack of] continuous care and other structural barriers to being as healthy as you possibly can be," Benatar said. "We can't really expect that just nine months of care is going to solve everybody's problems."

There has been some progress on this front, though. President Biden's American Rescue Plan Act of 2021 provided financial incentives to expand Medicaid and postpartum coverage to one year. While this expanded option initially went through just 2027, it was made permanent in Congress's Consolidated Appropriations Act of 2023. "In the past several years, the Biden administration has really invested in maternal health, equity issues, and disparity issues," said Dr. Hung.

While there are many policy solutions to solving maternal healthcare deserts in rural areas, women need access to resources now, and telemedicine may be the most efficient, realistic answer. Telemedicine provides a variety of health-related services through live video, remote patient monitoring, and electronic consultations. These services provide care, education, intervention, and monitoring assistance to patients and clinicians that are long-distance.

47% of healthcare workers plan to leave their jobs by 2025

Since 2003, the University of Arkansas for Medical Sciences has worked with a community of hospitals through broadband networks to provide services. This was initially in response to the poor health outcomes that have plagued the state for decades. In 2019, Arkansas was ranked number 49 regarding women and infant health outcomes, according to *America's Health Rankings*. Arkansas has also expanded use of the 24/7 obstetrical nurse call center model, which utilizes a staff of full-time nurses, patient service coordinators, and appointment center staff to talk with patients and provide guidelines that are evidence-based to support high-risk patients.

There are many initiatives that March of Dimes hosts involving telehealth. "Policy solutions around telehealth can address the limited access to maternity care in the U.S., expanding accessibility and providing more options for healthcare delivery," Stoneburner said. "The Tech to Save Moms Act will make a difference for families by investing in and promoting telehealth and digital tools that can be used by families in areas of the country with few or no maternity care providers. As a country, we need legislation like the Tech to Save Moms Act so that families do not need to extend themselves to access the high-quality maternity care they need and deserve."

Telemedicine, especially in predominantly rural states like Arkansas, can be a model for expanding access to more rural counties. But it is not always the most viable option to serve all rural communities—especially when many of them still struggle with internet access.

"The implementation is the fight," Dr. Robert Bullard noted. Investing in telehealth, expanding Medicaid, and creating more rural OB GYN programs like that at the University of Wisconsin is more vital than ever. There is no single solution to this nation-wide crisis. "We all have a role to play in prioritizing maternal health here in the U.S.," said Stoneburner. "Everyone can do their part so that all moms and babies have the best possible start, right in their local communities."

A Conversation with The Age of A.I.

Feud Over War Imperils Future Of

By SANTU
and JONAH E. B

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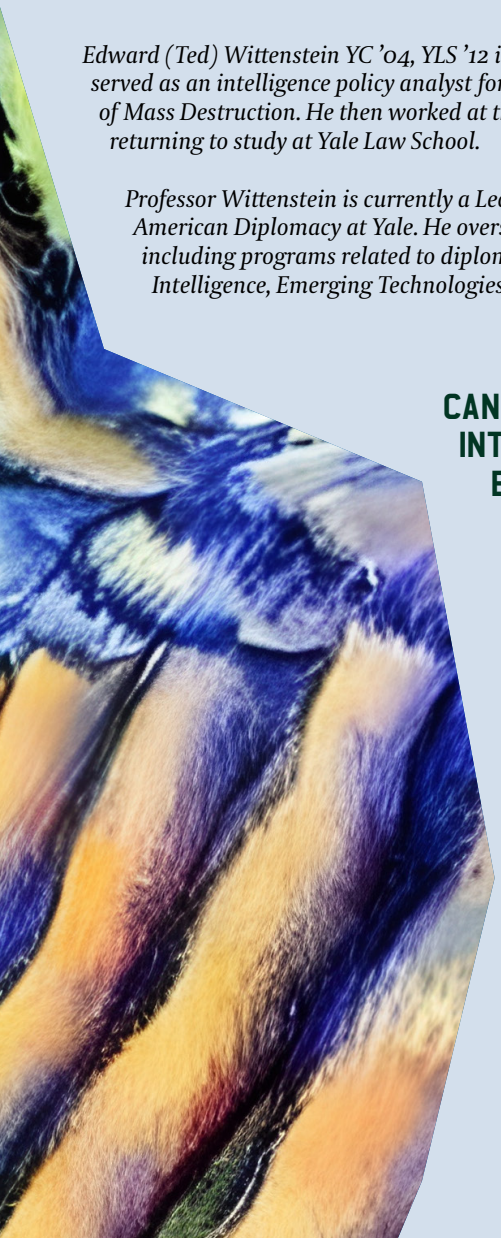
Ted Wittenstein

BY **KATE REYNOLDS**

Edward (Ted) Wittenstein YC '04, YLS '12 is a former intelligence professional and diplomat. After graduating from Yale College, he served as an intelligence policy analyst for the Commission on the Intelligence Capabilities of the United States Regarding Weapons of Mass Destruction. He then worked at the Office of the Director of National Intelligence and the U.S. Department of State, before returning to study at Yale Law School.

Professor Wittenstein is currently a Lecturer in Global Affairs and the Executive Director of the Johnson Center for the Study of American Diplomacy at Yale. He oversees the International Security Studies program at the Jackson School of Global Affairs, including programs related to diplomatic history, grand strategy, global security, and the Schmidt Program on Artificial Intelligence, Emerging Technologies, and National Power.

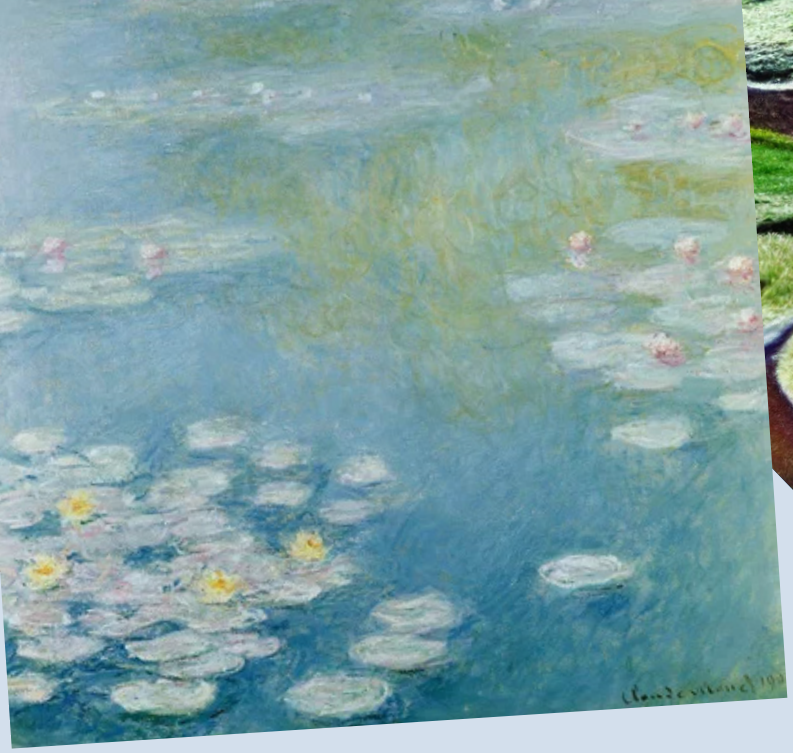


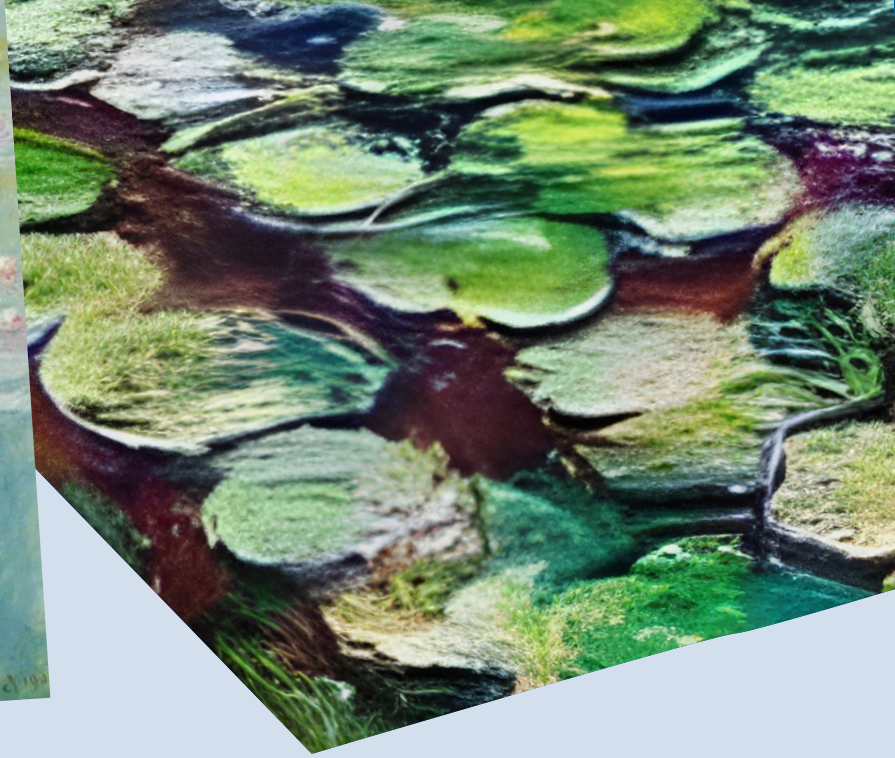
CAN YOU TELL US A BIT ABOUT YOUR EXPERIENCE BREAKING INTO AND WORKING IN INTELLIGENCE? HOW HAS THIS TRANSLATED OVER INTO YOUR WORK AT YALE?

With the Schmidt Program on Artificial Intelligence, Emerging Technologies and National Power, we are very excited about the opportunity to consider how trends in technology are impacting global affairs. This is something I worked on as an intelligence professional at a time when the intelligence community was grappling with trends in cybersecurity and emerging technology, which have really transformed the nature of analysis of global affairs.

My formative experiences in terms of intelligence issues were during the aftermath of 9/11 and the invasion of Iraq. I was a Yale undergrad back then, and it was a very awakening moment on campus for so many young people. I was not a global affairs-oriented student at that time—I was more STEM-oriented. So after 9/11, when everyone was looking for ways to get involved in public affairs, it became clear that my technology-proficient background was relevant to analyzing questions of weapons: nuclear, chemical, biological, and cyber.

I've always tried to bring this background into the classroom at Yale—how do you develop technical fluency among policy-oriented students? And the reverse—how do you expose our technical students to why their areas of research and focus are so relevant to the global affairs landscape?





CAN YOU DEFINE ARTIFICIAL INTELLIGENCE?

You cannot define AI. There are multiple definitions, and if you ask a computer scientist, a cognitive scientist, an ethicist, or a lawyer, they will give very different responses.

In my view, we're talking about highly sophisticated information processing, towards the frontier of autonomy, often involving algorithms, deep learning, and processing of big datasets—but there is no single way to define it.

This is something we talk about in our Schmidt program class—what is artificial? Currently, if you're following advances in synthetic biology, it looks like we're going to be able to program human cells as computers. Is that artificial? Is that biological? And when we say intelligence for a machine, why are we benchmarking machine performance against human capability? Is that the right framework? How do you understand a machine that's intelligent in certain functions—it can beat you at a complex game, but can't do first-grade math. Is that intelligent?

HOW WAS ARTIFICIAL INTELLIGENCE PART OF THE EQUATION IN INTELLIGENCE COLLECTION DURING 9/11 AND THE IRAQ WAR?

AI is always difficult to define. AI, as we currently see it, was not at all something that was on people's minds at the time of 9/11 and the Iraq War. I come from a generation where I remember the first time I saw a computer. I think there was certainly an understanding that technology and global communications needed to be more integral to how you thought about challenges and analyzed national security issues, but it wasn't really AI. No one was saying, at that time, "we have to try to automate the function or be able to do it at a scale you can now." I remember a time when people would say that Microsoft Excel was AI because you could automate a spreadsheet to do the sums or averages!

Since then, the technology has significantly advanced, but the human nature of analysis is something that is never going to go away. This is really a question of how you equip humans with tools, and partner with the machine as a team. The machines can help your work, but to be a human analyst requires a deeper understanding of global affairs, language expertise, functional expertise, cultural expertise—things that a machine is unlikely to replicate or replace.

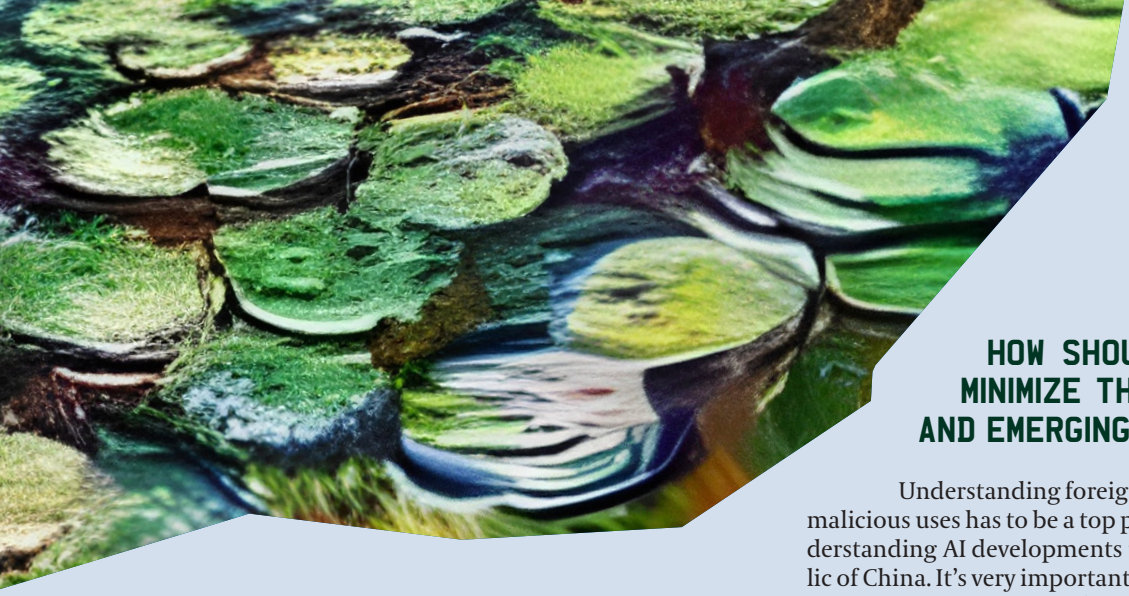
The current challenge facing the intelligence community is how much global information is already available through open sources. What does that mean for information that might be collected through any kind of clandestine means? For example, information on my phone about the war in Ukraine would never have been accessible to me as an intelligence analyst 15 years ago. I would have had to bury myself in a nondescript compartment to see the imagery analysis and read the field reports. In some ways, a lot of intelligence has been democratized. So, the question is, what is the value of intelligence? What makes something intelligence, as opposed to information? This is a real problem in the digital age. We have too much information, not too little.

HOW HAS THE DEMOCRATIZATION OF INTELLIGENCE COLLECTION IMPACTED THE INTELLIGENCE SPACE? WHAT ROLE DOES THE PRIVATE SECTOR PLAY IN INTELLIGENCE ANALYSIS?

Well, it means that your network of contacts and expertise needs to be more diverse, to include private sector actors and non-profit academic entities. These are individuals and organizations that have a lot of valuable insights.

Let's just take, for example, AI and emerging technologies—what are potential risks, areas of malicious use, or vulnerabilities? The expertise on that question likely does not reside in the government but rather in the U.S. private sector. That doesn't mean the government analyst can't analyze the question, but that they need to have relationships where they draw on and incorporate those expertise and insights to inform their thinking about world events.

I don't think the fundamental concepts are going away. To be an astute analyst you have to be able to see the world from different vantage points, challenge your own assumptions, communicate your confidence judgments clearly to policymakers who may not have time in their day, or don't want to hear bad or conflicting news.



WHAT ARE THE WAYS YOU CAN ENVISION AI BEING A THREAT OR DISADVANTAGE TO US?

AI is like most technology tools in that it is dual use, so it has both a beneficial as well as a potential nefarious use. It is a value neutral tool, so it is up to the developer to program in ethical restraints, and up to the user to be responsible. Ultimately, a determined malicious user will either undermine those constraints or develop a tool that doesn't have those constraints.

I think the most immediate concern with advances in natural language processing, like ChatGPT and GPT-4, is disinformation. These tools have the ability to automate human-looking text at scale, but also narrowly target text to you based on your social media presence or profile in a way that might convince you. So, the ability to create very sophisticated disinformation campaigns is a real area of concern.

The automation of weaponry is another area of significant concern. These tools are, again, not in themselves a concern—it depends on the intent of the attacker. For example, if you look at the way that Russia has conducted its humanitarian atrocity and genocide in Ukraine without autonomous weapons, it is reasonable to believe that they would use an autonomous tool indiscriminately in a way that's not aligned with our values.

In theory, an autonomous weapon might actually enhance your ability to adhere to law of war principles. It could limit collateral damage because it might be able to distinguish a military from a civilian target better than a human might be able to. But other countries might not have that same concern or might even be purposefully looking to target a school or a hospital. So, this is the future that we're headed for.

I think that the advances in drones that we've seen in the war in Ukraine present a rather destabilizing future in autonomous warfare. However, it is not truly autonomous in a sense that this technology takes the humans fully out of the loop. The humans are still involved. It is still the human making a decision. The challenge is what the U.S. Defense Department calls the question of "meaningful human control." So yes, you might have your finger on the trigger button, but what about all of the information that has gone into the decision up until that final point? What if all of that information is based on AI imagery, facial recognition, voice recognition, and geolocation? And, as the human, do you have a meaningful, real-time ability to explain, understand, or even question the analysis that will underpin your decision? So, at that point, it may not matter whether you're on the button or not. I think this is part of where we're headed in the future of conflict.

HOW SHOULD THE UNITED STATES MINIMIZE THE RISK OF MISUSE OF AI AND EMERGING TECHNOLOGIES?

Understanding foreign advancements in AI and potential malicious uses has to be a top priority of the U.S., which means understanding AI developments taking place in the People's Republic of China. It's very important to be sure the U.S. is not vulnerable to strategic surprise on this question. Now, there are potential misuses that can come from all parts of the world. I'm not saying you only focus on one, but I do think China has expressed public desire to use these tools for its own efforts at disinformation and narrative controlling, both within its own country and globally.

Another challenge is that you have to consistently analyze your own systems to identify their own vulnerabilities and how malicious actors might exploit them. Yesterday on campus we hosted Ram Kumar from Microsoft, who is running an AI Red Team to conduct adversarial machine learning research. The goal is to predict how other entities might try to undermine, exploit, or repurpose their tools. The challenge is that this type of research—although defensive in nature— produces insights that can be useful to another malicious actor. When Microsoft and the U.S. engage in that work, we know that it's defensive, but another country might conduct the same research to look for offensive ways to manipulate or undermine our own systems. The difficulty is deciphering intent between offensive and defensive. This ambiguity creates a volatile international system that you have to be very aware of.

HOW DO YOU PROGRAM AN ETHICAL SYSTEM INTO AI? BASED ON YOUR EXPERIENCE, HOW DOES ONE CONSTRUCT THIS SYSTEM OF ETHICS?

There are a number of questions embedded in that question. One is, what is your ethical code or constraints? The other would be, how would you build it into a machine? Then there is the broader question of AI safety and trustworthiness. Even if you created constraints in AI, how do you ensure they can't be manipulated or undermined? It's very difficult.

If you look at generative AI and large language models like ChatGPT, even with the constraints built in—you can't ask it to do something illegal, or use hateful or racist language—you still could get it to do that. It's not difficult to create the so-called "jailbreaks," or methods of subverting these ethical constraints. Malicious actors are thinking about and trying to do that all the time. Ethics are extremely important, but it's actually the safety architecture that embeds and protects the ethics from subversion.

I am delighted that Professor Luciano Floridi has joined Yale as the founding director of the new Digital Ethics Center. The Schmidt Program has partnered with Professor Floridi to develop a weekly Digital Ethics Workshop that examines these complex questions.

SHENZHEN REWIND

BY **LINXI CINDY ZENG**

For three years, Shenzhen was my address, but not my home. The miracle city only felt aggressively utilitarian and vacuously cold to me—until I started seeing it through color-suppressant filters. For me, it takes more than one story to understand a place, and fewer noises to come to appreciate it.

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